

A Unified Maximum-Likelihood Framework for Signal Reconstruction: From Optimal Filtering to Noise-Aware Linear Autoencoders

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Abstract

Signal reconstruction under structured measurement noise is a shared problem across signal processing, statistical latent-variable modeling, and representation learning, but these communities typically study it through separate estimator families. In this work we formulate optimal filtering (OF), expectation-maximization PCA (EM-PCA), and linear encoder-decoder models within a single Gaussian maximum-likelihood framework in which the noise covariance is explicitly incorporated into the objective. From this unified noise-aware loss, we derive classical OF as a rank-1 constrained estimator, show that EMPCA is its learned rank- k subspace generalization, and prove that a tied linear autoencoder in whitened space minimizes the same objective and therefore recovers the same signal subspace.

This yields a hierarchy of estimators ranging from basic OF to time-shift OF, joint-channel correlated OF, and learned low-rank EMPCA/linear-autoencoder reconstruction. The whitening transform gives the common representation-learning view: all methods become projections in a geometry induced by the noise model, and differ only by the structural constraint imposed on the reconstruction map. We further discuss convergence, model-order selection, and conditioning in practical

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EMPCA implementations. The framework is evaluated on cryogenic detector pulse data using both simulation and real K- α events, where we verify the rank-1 OF-EMPCA equivalence and study how reconstruction quality changes with subspace rank. The central machine-learning message is that, when the noise statistics are known or can be estimated, they should be encoded in the training objective rather than ignored by isotropic mean-squared error. This linear theory defines the baseline and motivation for nonlinear extensions developed in a companion paper.

1 Introduction

1.1 Three communities, one problem

Modern low-threshold experiments increasingly rely on reconstructing very small signals from noisy time series rather than simply counting threshold crossings. This is especially true in cryogenic detectors, where rare interactions must be identified from waveform traces whose amplitudes, arrival times, and shapes are all affected by the detector response and by the noise environment. In superfluid-helium concepts such as DELight, the deposited recoil energy is partitioned among several observable channels, including quasiparticles, UV photons, and long-lived excimers. For the quasiparticle branch, ballistic transport, quantum evaporation, calorimetric collection, and readout noise together determine the measured pulse trace. The practical inference problem is therefore not merely detection, but statistically efficient reconstruction of a signal family from colored, heteroskedastic, and potentially multi-channel noise.

This reconstruction problem is studied by at least three communities that use closely related ideas but different language. In signal processing, matched filters and optimal filters estimate amplitudes by projecting data onto a known template with inverse-noise weighting. In statistics, principal-component and subspace methods replace a fixed template by a learned low-dimensional signal model, with expectation-maximization PCA (EMPCA) providing a natural extension to heteroskedastic noise. In machine learning, encoder-decoder models and autoencoders recast reconstruction as representation learning in a latent space. All three communities address the same underlying task: recover structure from noisy observations by constraining reconstruction to a low-complexity model.

The commonality becomes particularly clear in the detector setting considered here. A single fixed pulse template is often an excellent first approximation, which makes optimal filtering a strong baseline. At the same time, realistic pulse families are broadened by timing jitter, geometry dependence, channel coupling, transport effects, and other sources of variation, so a rank-1 model is not always sufficient. This is precisely the regime in which learned subspaces and encoder–decoder viewpoints become useful. The central claim of this paper is that these are not competing ideas, but constrained versions of one noise-aware inference principle.

1.2 The core gap

Despite their close mathematical relationship, optimal filtering, EMPCA, and linear autoencoders are usually presented as separate methods. The signal-processing literature emphasizes estimator optimality for a fixed template under known noise; the statistical literature emphasizes low-rank structure and EM updates; and the machine-learning literature emphasizes latent representations and reconstruction losses. What is missing is a single formulation that makes their relationship explicit in the setting most relevant to scientific instrumentation: known or estimated noise covariance, structured waveform families, and the need to compare fixed-template and learned-subspace reconstruction on the same footing.

That gap matters both conceptually and practically. Conceptually, it obscures the fact that rank-1 optimal filtering, rank- k EMPCA, and noise-aware linear autoencoders solve the same Gaussian maximum-likelihood problem under different constraints on the admissible reconstruction. Practically, it makes it harder to transfer tools across communities: whitening and generalized least squares from signal processing, subspace learning from statistics, and representation-learning language from machine learning. In cryogenic pulse analysis this fragmentation is especially costly, because the relevant data combine colored noise, timing variation, and channel correlations while still benefiting from strong physical priors on the signal manifold.

This paper closes that gap for the linear regime. We formulate a unified noise-aware objective, show how classical optimal filtering appears as the rank-1 constrained case, show how EMPCA extends it to a learned rank- k subspace, and then show that the same objective can be written in encoder–decoder form as a noise-aware tied linear autoencoder in whitened space. In this view, the key modeling choice is not whether one uses a filter, a subspace

method, or an autoencoder, but which reconstruction constraint is imposed and how the noise geometry is encoded in the loss.

1.3 Related work and positioning

Signal processing: optimal filtering and matched filtering. The optimal filter (OF) for a known deterministic signal in Gaussian noise is a classical result, derived as the minimum-variance unbiased estimator via the Gauss–Markov theorem or, equivalently, as the maximum-likelihood estimator for the Gaussian model [1]. In the signal-processing literature it appears as the matched filter [2], as the Wiener filter for mean-square-error minimization [3], and as the generalized least-squares (GLS) estimator in the regression literature [4]. For calorimetric detectors specifically, OF was introduced by (author?) [5] and is reviewed in (author?) [6] as the standard amplitude estimator for cryogenic sensor readout. The present paper places OF explicitly within a Gaussian ML framework, derives its Cramér–Rao bound, and identifies it as the rank-1 special case of a more general constrained ML family.

Statistics: probabilistic PCA, factor analysis, and EMPCA. Probabilistic PCA (PPCA) [7] and factor analysis [8] are generative latent-variable models that perform low-rank signal recovery under Gaussian noise. PPCA assumes isotropic noise ($\Sigma = \sigma^2 I$), recovering standard PCA as its ML estimate. Factor analysis relaxes this to diagonal but anisotropic noise ($\Sigma = \text{diag}(\psi_1, \dots, \psi_d)$) and is equivalent to heteroskedastic PCA in one-dimensional feature directions. (author?) [9] introduced EMPCA as an EM algorithm for PCA under per-observation heteroskedastic weights, and (author?) [10] applied a frequency-domain variant to microcalorimeter pulse data. The present paper goes beyond these works in three directions: we derive EMPCA as the rank- k constrained ML estimator for a fully general covariance Σ , we prove the exact algebraic equivalence between rank-1 EMPCA and the classical OF estimator, and we connect EMPCA to a noise-aware linear autoencoder via the Bridge Theorem.

Machine learning: linear autoencoders and noise-aware losses. Linear autoencoders with tied weights recover the PCA subspace when trained with mean-squared error (MSE) loss [11]. The connection between PCA and

autoencoders has been revisited repeatedly [12, 13], and the equivalence between linear AEs and PCA in the over-complete and tied settings is well established. Noise-aware or weighted reconstruction losses appear in the denoising AE literature [14] and in heteroskedastic regression [15], but the explicit connection to weighted PCA and EMPCA has not been formalized. The Bridge Theorem in this paper provides that connection: a tied linear AE with a noise-covariance-weighted reconstruction loss minimizes the same objective as EMPCA and therefore recovers the same signal subspace. This provides the principled baseline for nonlinear extensions developed in the companion paper.

Scope of the present paper. We restrict attention to linear subspace models, known or estimated noise covariance, and single-detector waveform reconstruction. We do not treat nonlinear manifold learning, deep architectures, variational inference, or online/streaming variants of the algorithms, which are reserved for the companion paper.

1.4 Contributions

The contributions of this paper are as follows.

1. We formulate optimal filtering, EMPCA, and linear encoder–decoder reconstruction within a single Gaussian maximum-likelihood objective with explicit covariance weighting. This gives a common noise-aware foundation for all later sections.
2. We prove in Theorem 4.1 that, under matched preprocessing, whitening, and weighting assumptions, rank-1 EMPCA is equivalent to classical optimal filtering. We further state the exact algebraic degeneracy in Theorem 4.2.
3. We organize basic optimal filtering, time-shift optimal filtering, joint-channel correlated optimal filtering, and rank- k EMPCA into a hierarchy of constrained estimators. This makes precise how fixed-template, manifold, and learned-subspace reconstruction fit within the same weighted projection framework.
4. We show in Section 5.3 that a noise-aware tied linear autoencoder in whitened space minimizes the same objective as EMPCA, so the

encoder–decoder formulation is not an unrelated add-on but the natural endpoint of the linear theory.

5. We discuss convergence, conditioning, and model-order selection for practical EMPCA implementations, and we validate the framework on cryogenic detector pulse data using both simulation and real K- α events.

The scope of this paper is deliberately limited to linear subspace models with a Gaussian noise-aware objective. We do not attempt here to develop nonlinear triggering or nonlinear autoencoder architectures. Instead, the purpose of the present manuscript is to establish the linear theory cleanly enough that nonlinear extensions can later be judged against a principled baseline rather than against an arbitrary collection of heuristics.

1.5 Paper organization

Section 2 introduces the unified noise-aware objective, the whitening transform, and the estimator hierarchy. Section 3 develops optimal filtering as the rank-1 constrained case, including time-shift and joint-channel extensions. Section 4 develops EMPCA as the rank- k constrained case and proves its rank-1 equivalence to optimal filtering. Section 5 reframes the same objective in encoder–decoder language and connects EMPCA to a noise-aware linear autoencoder. Section 6 discusses convergence, initialization, model-order selection, and conditioning. Section 7 provides experimental validation on cryogenic detector pulse data. Section 8 discusses the broader significance of noise-aware losses, the relation to probabilistic PCA and latent-variable modeling, and the limitations that motivate a companion nonlinear paper. Technical proof details and detector-specific simulation material are collected in the appendix.

2 The Unified Noise-Aware Objective

This section reformulates Optimal Filtering (OF) and Expectation–Maximization Principal Component Analysis (EMPCA) within a single maximum-likelihood (ML) framework. The main goal is to show that these methods are not fundamentally different estimators, but rather different constrained solutions of

the same Gaussian inverse problem. In particular, we distinguish three levels of equivalence:

1. **Estimator equivalence:** two estimators produce the same scalar amplitude estimate under matched preprocessing and weighting;
2. **Subspace equivalence:** two methods recover the same signal subspace, up to a trivial gauge freedom (sign, phase, or invertible basis transformation);
3. **Amplitude equivalence:** once projected onto the same subspace with consistent normalization, the reconstructed coefficients agree.

Under appropriate assumptions, rank-1 EMPCA degenerates exactly to the basic optimal filter. More generally, OF with time shift and joint-channel correlated OF can also be understood as special cases of the same weighted projection principle. EMPCA extends this picture by replacing a fixed one-dimensional template subspace with a learned low-dimensional signal subspace.

2.1 Generative model

We explicitly separate indices as follows:

- event index: $i = 1, \dots, N$,
- feature index: $j = 1, \dots, d$,
- channel index: $c = 1, \dots, C$,
- frequency index: k when working in Fourier space.

We first treat the single-channel case and suppress the channel index. For each event i , we observe a vector

$$x_i \in \mathbb{C}^d,$$

representing either a time-domain trace or frequency-domain coefficients. We assume the generative model

$$x_i = a_i s + n_i, \tag{1}$$

where

- $s \in \mathbb{C}^d$ is a fixed signal template,
- $a_i \in \mathbb{R}$ (or $\mathbb{C}$ in frequency-domain form) is the event amplitude,
- $n_i \sim \mathcal{N}(0, \Sigma)$ is zero-mean Gaussian noise with known covariance $\Sigma \in \mathbb{C}^{d \times d}$.

Define the whitening transform

$$\tilde{x}_i = \Sigma^{-1/2} x_i, \quad \tilde{s} = \Sigma^{-1/2} s. \quad (2)$$

Then in whitened coordinates,

$$\tilde{x}_i = a_i \tilde{s} + \tilde{n}_i, \quad \tilde{n}_i \sim \mathcal{N}(0, I). \quad (3)$$

Stacking events row-wise gives the whitened data matrix

$$\tilde{X} = \begin{bmatrix} \tilde{x}_1^\dagger \\ \vdots \\ \tilde{x}_N^\dagger \end{bmatrix} \in \mathbb{C}^{N \times d}.$$

The advantage of (2)–(3) is that both OF and EMPCA become weighted least-squares projections in a common Euclidean geometry.

2.2 The Gaussian ML objective

Let $x \in \mathbb{C}^d$ denote one trace in either time- or frequency-domain representation, with noise model

$$x = \hat{s} + n, \quad n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma), \quad \Sigma \succ 0. \quad (4)$$

The Gaussian negative log-likelihood, up to additive constants, is

$$\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s}). \quad (5)$$

This immediately yields a unifying interpretation:

- **Optimal Filtering** constrains the reconstructed signal to lie in a one-dimensional template subspace;
- **EMPCA** constrains the reconstructed signal to lie in a learned k -dimensional subspace.

Thus both methods solve the same weighted ML problem, differing only in the admissible signal model.

2.2.1 Optimal Filtering as a rank-1 constrained ML estimator

For a known template s_0 and unknown scalar amplitude A , the signal model is

$$\hat{s} = A s_0.$$

Substituting into (5) and minimizing with respect to A gives the generalized least-squares estimator

$$\hat{A}_{\text{OF}} = \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0}. \quad (6)$$

This is exactly the optimal filter estimator written in covariance form. For stationary noise in Fourier space, Σ is diagonal and (6) reduces to the usual PSD-weighted OF formula.

2.2.2 EMPCA as a subspace-constrained ML estimator

Let $U = [u_1, \dots, u_k] \in \mathbb{C}^{d \times k}$ denote a basis for a k -dimensional signal subspace. EMPCA models the reconstructed signal as

$$\hat{s} = U c,$$

where $c \in \mathbb{C}^k$ is the coefficient vector for the given event. Then (5) becomes

$$\chi^2(c) = (x - U c)^\dagger \Sigma^{-1} (x - U c). \quad (7)$$

Minimization yields the normal equations

$$U^\dagger \Sigma^{-1} U c = U^\dagger \Sigma^{-1} x,$$

hence

$$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x. \quad (8)$$

Therefore the reconstructed signal is

$$\hat{s}_{\text{EMPCA}} = U \hat{c}. \quad (9)$$

Equation (8) makes the connection to OF explicit: OF is simply the special case $k = 1$ with $U = s_0$.

2.3 Whitening transform

The whitening transform $\Sigma^{-1/2}$ was introduced formally in (2). Here we discuss its practical construction, its geometric interpretation, and the numerical considerations that arise in detector applications.

Geometric role. Whitening converts the non-Euclidean Σ^{-1} -weighted inner product into a standard Euclidean inner product. For any two vectors $u, v \in \mathbb{C}^d$,

$$u^\dagger \Sigma^{-1} v = (\Sigma^{-1/2} u)^\dagger (\Sigma^{-1/2} v) = \tilde{u}^\dagger \tilde{v}. \quad (10)$$

After whitening, the noise covariance is the identity, and all Σ^{-1} -weighted projections become ordinary orthogonal projections. This is the key simplification that makes the geometric interpretation of Section 4.5 rigorous.

Stationary single-channel case: diagonal PSD. When the noise is wide-sense stationary, the covariance matrix Σ is circulant and is diagonalized by the DFT matrix F :

$$\Sigma = F^\dagger \text{diag}(J_1, \dots, J_d) F, \quad (11)$$

where $J_k = \mathbb{E}[|N_k|^2]$ is the one-sided noise power spectral density (PSD) at frequency bin k . The whitening transform is then

$$\Sigma^{-1/2} = F^\dagger \text{diag}(J_1^{-1/2}, \dots, J_d^{-1/2}) F, \quad (12)$$

and is applied in the frequency domain by element-wise division:

$$\tilde{X}_k = \frac{X_k}{\sqrt{J_k}}. \quad (13)$$

This is the most common implementation in detector analysis: whiten in Fourier space by dividing each frequency bin by the square root of the measured PSD.

General case: eigendecomposition and Cholesky. When Σ is not circulant (e.g. for short traces, non-stationary noise, or multi-channel data with cross-channel correlations), two approaches are standard.

Eigendecomposition. Write $\Sigma = V \Lambda V^\dagger$ where $V \in \mathbb{C}^{d \times d}$ is unitary and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $\lambda_j > 0$. Then

$$\Sigma^{-1/2} = V \Lambda^{-1/2} V^\dagger. \quad (14)$$

This is numerically stable but requires $O(d^3)$ work to compute.

Cholesky. Write $\Sigma = L L^\dagger$ where L is lower triangular. Then $\Sigma^{-1} = L^{-\dagger} L^{-1}$ and $\Sigma^{-1/2} = L^{-\dagger}$ (up to a choice of square root convention). Cholesky is faster for dense matrices and numerically preferred when Σ is well-conditioned.

Numerical stability: regularization and frequency masking. In practice J_k may be very small at certain frequency bins (e.g. above the signal bandwidth or below the low-frequency floor). Dividing by $\sqrt{J_k}$ amplifies noise at those bins and can destabilize subsequent computations. Two standard remedies are:

1. *Ridge regularization.* Replace $J_k \rightarrow J_k + \varepsilon$ for a floor $\varepsilon > 0$ chosen as a small fraction of $\max_k J_k$ or as the numerical noise floor. This biases the whitened inner product slightly but ensures bounded amplification.
2. *Frequency masking.* Restrict the analysis to a frequency band $[f_{\min}, f_{\max}]$ where the signal has appreciable power relative to the noise. Bins outside the band are set to zero before whitening. This is equivalent to projecting both signal and noise onto the informative subspace and is the standard approach in cryogenic detector optimal filtering.

Connection to the estimator hierarchy. Once whitened, every estimator in Table 1 performs an orthogonal projection in $\mathbb{C}^d$ with the standard Euclidean norm. The structure of the projection (rank-1 fixed, rank-1 manifold, or rank- k learned) is what distinguishes the estimators, not the geometry of the inner product. This separation of noise model (captured by Σ and the whitening) from signal model (captured by the constraint) is the central organizing principle of the framework.

2.4 Three constraints, three estimators

The analysis of this section can be summarized as follows:

1. **Basic OF** is the ML projection onto a fixed rank-1 template subspace.
2. **OF with time shift** is the ML projection onto a nonlinear one-parameter shift manifold generated by a common template.
3. **Joint-channel correlated OF** is the same weighted ML projection principle applied to a stacked multi-channel vector with full covariance weighting.
4. **EMPCA** is the corresponding learned low-rank generalization, where the signal is reconstructed in a data-driven subspace rather than a fixed template direction.

Table 1: Hierarchy of constrained ML estimators under the unified noise-aware objective. All methods minimize $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$. “DoF” counts the real degrees of freedom in the reconstruction map. “Degeneracy condition” states when the row above and below coincide exactly.

Estimator	Constraint on $\hat{s}$	Subspace dim.	Solution form	Degeneracy condition
Basic OF	$\hat{s} = A s_0$, s_0 fixed	$k = 1$ (fixed)	$\hat{A} = (s_0^\dagger \Sigma^{-1} x) / (s_0^\dagger \Sigma^{-1} s_0)$	Signal is exactly $\propto s_0$
Time-shift OF	$\hat{s} = A e^{-i\omega t_0} s_0$, t_0 unknown	$k = 1$ (shift manifold)	$\hat{t}_0 = \arg \max_{t_0} \hat{A}(t_0) $	No timing jitter; collapses to basic OF
Joint-channel OF	$\hat{s} = A s$, $s \in \mathbb{C}^M$ joint template	$k = 1$ (joint)	$\hat{A} = (s^\dagger \Sigma^{-1} x) / (s^\dagger \Sigma^{-1} s)$	Channels uncorrelated; factorizes per channel
Rank- k EMPCA	$\hat{s} = U c$, $U \in \mathbb{C}^{d \times k}$ learned	k (learned)	$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x$	$k = 1$, $\text{span}(U) = \text{span}(s_0)$: equals basic OF
Noise-aware linear AE	$\hat{s} = P P^\dagger \Sigma^{-1} x$, $P^\dagger \Sigma^{-1} P = I$	k (learned)	P^* = top- k eigenvectors of $\hat{R}_x \Sigma^{-1}$	Equivalent to EMPCA at convergence (Theorem 5.2)

Therefore OF and EMPCA should be viewed not as competing estimators, but as different constrained solutions of the same Gaussian inverse problem. Exact degeneracy occurs when the learned EMPCA subspace collapses to the relevant OF template subspace; improvement occurs only when the true signal family has additional dimensions that a fixed-template OF cannot capture.

Table 1 summarizes the full estimator family. Each row is a constrained minimizer of the same Gaussian ML objective (5); the rows differ only in how the admissible reconstruction is constrained.

Three structural observations follow from Table 1. First, the constraint complexity increases monotonically from top to bottom: basic OF is the most constrained (one free parameter, fixed geometry) and the noise-aware linear AE is the least constrained (a full k -dimensional subspace learned from data). Second, the noise model Σ enters identically in every row through the same quadratic form — the methods differ in their signal model, not in how they handle noise. Third, degeneracy propagates downward: each row collapses to the row above it when the learned subspace converges to the fixed template direction. Sections 3–5 prove these degeneracies formally.

2.5 Multi-channel extension

Many physical detectors record multiple channels simultaneously. The framework extends naturally to this case by stacking channels into a joint observation vector.

Joint observation model. For C channels and a trace length of d samples per channel, define the stacked data vector

$$x = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \quad s = \begin{bmatrix} s^{(1)} \\ \vdots \\ s^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \quad (15)$$

and model the noise as

$$n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma_{\text{joint}}), \quad (16)$$

where $\Sigma_{\text{joint}} \in \mathbb{C}^{Cd \times Cd}$ has the block structure

$$\Sigma_{\text{joint}} = \begin{bmatrix} \Sigma^{(11)} & \Sigma^{(12)} & \dots & \Sigma^{(1C)} \\ \Sigma^{(21)} & \Sigma^{(22)} & \dots & \Sigma^{(2C)} \\ \vdots & & \ddots & \vdots \\ \Sigma^{(C1)} & \dots & \dots & \Sigma^{(CC)} \end{bmatrix}. \quad (17)$$

Here $\Sigma^{(cc)} \in \mathbb{C}^{d \times d}$ is the intra-channel covariance and $\Sigma^{(cc')} \in \mathbb{C}^{d \times d}$ captures the cross-channel noise correlation between channels c and c' . For stationary noise, $\Sigma^{(cc)}$ is circulant and $\Sigma^{(cc')}$ is the cross-spectral density matrix.

Unified objective in the multi-channel case. The Gaussian ML objective is unchanged in form:

$$\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma_{\text{joint}}^{-1} (x - \hat{s}). \quad (18)$$

Substituting the rank-1 constraint $\hat{s} = As$ gives the joint-channel OF estimator (recovered in Section 3.3.1). Substituting the rank- k constraint $\hat{s} = Uc$ with $U \in \mathbb{C}^{Cd \times k}$ gives multi-channel EMPCA. The only change relative to the single-channel case is the dimension of the ambient space and the block structure of Σ_{joint} .

Whitening in the multi-channel case. The whitening transform is $\tilde{x} = \Sigma_{\text{joint}}^{-1/2} x$, computed via Cholesky or eigendecomposition of the joint covariance. In the frequency domain, when noise is stationary and cross-channel correlations are captured by a per-frequency cross-spectral matrix $\Sigma_f \in \mathbb{C}^{C \times C}$ for each bin f , the whitening decomposes into independent $C \times C$ Cholesky factors at each frequency, reducing to C scalar divisions per channel when channels are uncorrelated.

Reduction to single-channel case. If $\Sigma^{(cc')} = 0$ for $c \neq c'$ (no cross-channel noise correlation) and each $\Sigma^{(cc)} = \text{diag}(J_k^{(c)})$ is diagonal, the joint objective decouples into C independent single-channel objectives, one per channel. In this limit, joint-channel OF reduces to the arithmetic mean of per-channel OF estimates weighted by their respective Fisher informations, and multi-channel EMPCA reduces to independent single-channel EMPCA applied to the whitened channels simultaneously.

3 Optimal Filtering as Rank-1 Constrained ML

3.1 The rank-1 constraint and OF estimator

Specializing the unified objective (5) to the rank-1 constraint $\hat{s} = AS$ and the stationary single-channel case where Σ is diagonal with entries $J_k = \mathbb{E}[|N_k|^2]$ (the noise power spectral density at frequency bin k), the objective (5) becomes

$$\chi^2(A) = \sum_k \frac{|X_k - AS_k|^2}{J_k}, \quad (19)$$

which is the discrete Gaussian log-likelihood for uncorrelated frequency bins [1]. Setting $\partial\chi^2/\partial A^* = 0$ (Wirtinger derivative) gives the normal equation

$$\sum_k \frac{S_k^* X_k}{J_k} = \hat{A} \sum_k \frac{|S_k|^2}{J_k}, \quad (20)$$

hence

$$\hat{A} = \frac{\sum_k \frac{S_k^* X_k}{J_k}}{\sum_k \frac{|S_k|^2}{J_k}}. \quad (21)$$

This is the covariance-form estimator (6) in frequency coordinates. Define the filter kernel and normalization

$$\Phi_k = \frac{S_k^*}{J_k}, \quad N_\Phi = \sum_k \frac{|S_k|^2}{J_k}, \quad \Phi_k^{(0)} = \frac{\Phi_k}{N_\Phi}, \quad (22)$$

so that $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ is the filtered inner product of data with the normalized kernel — the discrete matched filter [2]. If A is constrained to be real, take $\text{Re } \hat{A}$ after filtering.

3.2 Estimator variance, Cramér–Rao bound, and energy resolution

The amplitude estimator $\hat{A}$ is unbiased and achieves the Cramér–Rao lower bound (CRB) for the problem, making it the minimum-variance unbiased estimator (MVUE) in the Gaussian model.

Unbiasedness. Taking the expectation of $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ under the model $X_k = AS_k + N_k$:

$$\mathbb{E}[\hat{A}] = \sum_k \Phi_k^{(0)} AS_k = A \sum_k \frac{S_k^* S_k}{J_k N_\Phi} = A \cdot \frac{N_\Phi}{N_\Phi} = A. \quad (23)$$

Estimator variance. Because the noise bins N_k are independent with $\mathbb{E}[|N_k|^2] = J_k$,

$$\text{Var}(\hat{A}) = \sum_k |\Phi_k^{(0)}|^2 J_k = \sum_k \frac{|S_k|^2}{J_k^2 N_\Phi^2} J_k = \frac{1}{N_\Phi^2} \sum_k \frac{|S_k|^2}{J_k} = \frac{1}{N_\Phi}. \quad (24)$$

Hence the standard deviation of the amplitude estimate is

$$\sigma_A = \frac{1}{\sqrt{N_\Phi}}, \quad N_\Phi = \sum_k \frac{|S_k|^2}{J_k}. \quad (25)$$

Cramér–Rao bound. The Fisher information for the scalar parameter A from a single trace is

$$\mathcal{I}(A) = -\mathbb{E} \left[\frac{\partial^2 \log \mathcal{L}}{\partial A^2} \right] = 2 \sum_k \frac{|S_k|^2}{J_k} = 2N_\Phi, \quad (26)$$

where the factor of 2 arises from the real–imaginary decomposition under the complex Gaussian model (or is absent if a real-valued model is used; conventions can differ by a factor of 2 depending on whether one-sided or two-sided PSD is used — see Appendix D). The CRB is $\text{Var}(\hat{A}) \geq 1/\mathcal{I}(A)$, and equality holds for $\hat{A}$, confirming it is efficient.

Energy resolution. Suppose amplitude A is proportional to the deposited energy E : $A = E/E_0$ for a known calibration energy E_0 (e.g. from a K- α X-ray). Then the baseline energy resolution (one-sigma width) is

$$\sigma_E = E_0 \cdot \sigma_A = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k} \right)^{-1/2}. \quad (27)$$

Equation (27) makes the design intuition explicit: σ_E decreases when the signal power $|S_k|^2$ is large at frequencies where the noise PSD J_k is small, i.e. where the detector is sensitive. The quantity N_Φ is therefore both the filter normalization and a direct measure of the signal-to-noise budget available to the OF estimator. Study B in Section 7.4 uses (27) to rank noise sources by their contribution to σ_E .

Implications for the rank- k extension. In whitened coordinates, the OF projector captures a fraction $N_\Phi/\text{Tr}(\Sigma^{-1}\hat{R}_x)$ of the total signal-to-noise budget. When the true signal family is not one-dimensional (e.g. timing jitter, pulse-shape variation, channel coupling), the remaining SNR is distributed across additional directions. EMPCA with $k > 1$ recovers those directions, yielding a lower residual χ^2 and, where applicable, a better energy estimate. This motivates the rank- k extension developed in Section 4.

3.3 Extensions: time-shift OF and joint-channel OF

Following Kurinsky Appendix E.2, include an unknown arrival-time shift t_0 via

$$S_k(t_0) = e^{-i\omega_k t_0} S_k, \quad X_k = A S_k(t_0) + N_k. \quad (28)$$

The weighted objective is

$$\chi^2(A, t_0) = \sum_k \frac{|X_k - A e^{-i\omega_k t_0} S_k|^2}{J_k}. \quad (29)$$

For fixed t_0 , the stationarity equation $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A}(t_0) = \frac{\sum_k \frac{e^{i\omega_k t_0} S_k^* X_k}{J_k}}{\sum_k \frac{|S_k|^2}{J_k}} = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k. \quad (30)$$

Therefore $\hat{A}(t_0)$ is the inverse DFT of the filtered spectrum, matching the Kurinsky E.2 interpretation.

To profile out amplitude, define

$$\chi_0^2 = \sum_k \frac{|X_k|^2}{J_k}, \quad B(t_0) = \sum_k \frac{e^{i\omega_k t_0} S_k^* X_k}{J_k}. \quad (31)$$

Then

$$\chi^2(A, t_0) = \chi_0^2 - A^* B(t_0) - AB^*(t_0) + |A|^2 N_\Phi. \quad (32)$$

At $A = \hat{A}(t_0) = B(t_0)/N_\Phi$,

$$\chi^2(t_0) = \chi_0^2 - \frac{|B(t_0)|^2}{N_\Phi} = \chi_0^2 - |\hat{A}(t_0)|^2 N_\Phi. \quad (33)$$

Hence the optimal delay is

$$\hat{t}_0 = \arg \min_{t_0} \chi^2(t_0) = \arg \max_{t_0} |\hat{A}(t_0)|. \quad (34)$$

For real-valued traces with Hermitian-symmetric spectra, this reduces to the usual “maximize filtered amplitude in time” criterion used by Kurinsky.

3.3.1 Joint Channel Correlated Optimum Filter

The same logic extends to multiple channels. Let the full data vector be formed by stacking all channel traces:

$$x = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(C)} \end{bmatrix}, \quad s = \begin{bmatrix} s^{(1)} \\ \vdots \\ s^{(C)} \end{bmatrix}.$$

Assume zero-mean Gaussian noise with full covariance

$$n \sim \mathcal{N}(0, \Sigma),$$

where Σ now contains both intra-channel and cross-channel correlations.

The joint-channel correlated optimal filter is exactly the generalized least-squares estimator

$$\hat{A}_{\text{JOF}} = \frac{s^\dagger \Sigma^{-1} x}{s^\dagger \Sigma^{-1} s}. \quad (35)$$

But this is algebraically identical to the rank-1 weighted subspace estimator obtained from (8) with $U = s$. Thus we have:

Proposition 3.1 (Degeneracy to joint-channel correlated OF). *Weighted EMPCA with the full inverse covariance Σ^{-1} is the multi-channel subspace generalization of joint-channel correlated optimal filtering. In the rank-1 case, when the EMPCA basis spans the joint template direction, weighted EMPCA reduces exactly to joint-channel correlated OF.*

Hence the role of EMPCA is not to change the underlying likelihood, but to replace a fixed joint template vector by a learned joint signal subspace. The noise correlations are handled identically in both methods through the same quadratic form $x^\dagger \Sigma^{-1} x$.

3.3.2 Time-Domain OF: Matched Filtering

A convolution kernel sliding over the time-domain trace.

Application: FIR filter We now turn to OF with an unknown arrival time. In frequency space, the shifted template is

$$S_k(t_0) = e^{-i\omega_k t_0} S_k, \quad X_k = AS_k(t_0) + N_k. \quad (36)$$

The weighted objective is

$$\chi^2(A, t_0) = \sum_k \frac{|X_k - Ae^{-i\omega_k t_0} S_k|^2}{J_k}. \quad (37)$$

For fixed t_0 , the amplitude estimator is

$$\hat{A}(t_0) = \frac{\sum_k e^{i\omega_k t_0} S_k^* X_k / J_k}{\sum_k |S_k|^2 / J_k}. \quad (38)$$

The optimal delay is then

$$\hat{t}_0 = \arg \max_{t_0} |\hat{A}(t_0)|. \quad (39)$$

The key point is that the shifted templates

$$\mathcal{M} = \{e^{-i\omega t_0} s : t_0 \in \mathbb{R}\}$$

form a one-parameter signal manifold rather than a single fixed vector. OF with time shift performs ML estimation over this manifold.

EMPCA enters when this manifold is approximated by a low-dimensional learned subspace. For small shifts, a Taylor expansion gives

$$e^{-i\omega t_0} s = s - i\omega t_0 s - \frac{1}{2}(\omega t_0)^2 s + \dots \quad (40)$$

Therefore the family of shifted pulses is approximately contained in the subspace spanned by

$$\{s, \omega s, \omega^2 s, \dots\},$$

or, in time-domain notation, by

$$\{s, \partial_t s, \partial_t^2 s, \dots\}.$$

This yields the following interpretation:

Proposition 3.2 (Degeneracy to OF with shift). *OF with time shift is the ML estimator obtained by restricting the signal model to the nonlinear shift manifold $\mathcal{M}$. EMPCA degenerates to shifted OF when the learned EMPCA subspace is exactly the span of the shift family; more generally, low-rank EMPCA provides a linear-subspace approximation to shifted OF when the dominant variation in the data is timing jitter around a common template.*

In this sense, basic OF, shifted OF, and rank- k EMPCA form a nested hierarchy:

basic OF $\subset$ shifted OF $\subset$ low-rank EMPCA approximation to shift variation.

3.4 Numerical verification

Using the strict real-data notebook setup with identical baseline correction, identical PSD weighting, and matched train/test conventions, the equivalence between OF and rank-1 EMPCA was checked numerically on real K- α data.

These results provide strong empirical support that OF and rank-1 EMPCA are functionally equivalent on this dataset once the assumptions of Theorem 4.1 are enforced. In particular:

- the weighted subspace cosine confirms that the learned rank-1 EMPCA direction is essentially identical to the OF template direction in whitened space;

Table 2: Real-data OF vs. rank-1 EMPCA verification metrics under matched preprocessing and weighting.

Metric	Value
$\max a_{\text{OF}} - a_{\text{template GLS}} $	5.5339×10^{-4}
Weighted subspace cosine $\rho_w(U, s)$	0.9999999655
Amplitude correlation r	0.9942010763
Median relative error (origin-fit)	9.6782×10^{-5}
Residual KS statistic D	0.0302891
Residual KS p -value	0.2702893

- the very small relative error confirms amplitude equivalence up to numerical precision and normalization convention;
- the compatible residual distributions indicate that both methods yield the same effective fit quality once projected onto the same one-dimensional signal model.

3.5 Limitation: the fixed template assumption

The OF derivation above assumes that every signal in the dataset is a scalar multiple of the *same* known template S_k . This is a strong constraint that breaks down in several realistic situations.

Timing jitter. If pulses arrive at unknown times $t_{0,i}$, the true signal for event i is $e^{-i\omega_k t_{0,i}} S_k$, not S_k . Time-shift OF (Section 3.3.2) mitigates this by searching over t_0 , but still restricts the reconstruction to a one-dimensional manifold parametrized by a single delay. When the jitter distribution is broad or the template has complex time structure, a fixed-template approach assigns incorrect weights across frequencies and under-estimates the total signal variance.

Pulse-shape variation. Physical processes such as position-dependent detector response, geometry inhomogeneity, channel cross-talk, and saturation effects cause the pulse shape itself to vary from event to event. If the true signal family spans a k -dimensional subspace (e.g. one axis for the mean pulse, one for a symmetric broadening mode, one for a tilt or asymmetric component), then OF — constrained to project onto a single fixed direction

— is *subspace-mismatched*: it conflates multiple physical modes into a single amplitude and cannot resolve them.

Bias and efficiency loss. Formally, when s_0 is not aligned with the true signal s , the OF estimator satisfies

$$\mathbb{E}[\hat{A}_{\text{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0} \neq A \quad (\text{if } s_0 \not\propto s). \quad (41)$$

The estimator is biased toward zero in proportion to the mismatch angle in whitened space. Additionally, the variance formula (24) is no longer the CRB for the true signal; the effective SNR loss is $(s_0^\dagger \Sigma^{-1} s)^2 / [(s_0^\dagger \Sigma^{-1} s_0)(s^\dagger \Sigma^{-1} s)]$, which equals $\cos^2 \theta_w$, the squared cosine of the angle between s_0 and s in whitened space.

What rank- k EMPCA provides. EMPCA removes the fixed-template assumption by replacing s_0 with a *learned* k -dimensional subspace $\text{span}(U)$. When the true signal family lies within this subspace, the EMPCA estimator is unbiased and achieves the CRB for the subspace-constrained problem. The cost is that the subspace must be estimated from data, introducing a finite-sample estimation error analyzed in Section 4. The benefit is a strictly lower residual χ^2 (Proposition 4.4) and, for structured signal families, a significant gain in effective SNR.

4 EMPCA as Rank- k Constrained ML

4.1 The rank- k constraint

Motivation. The optimal filter of Section 3 assumes that every signal in the dataset is a scalar multiple of a fixed, known template s_0 . This is adequate when the signal family is truly one-dimensional — that is, when pulse shape, amplitude, and timing are all fixed and the only unknown is the overall scale A . In practice, however, real signal families have more structure: pulses vary in shape due to position dependence of the detector response, timing jitter introduces a continuum of shifted versions of the nominal template, and multi-mode excitations can populate several orthogonal directions simultaneously. A more flexible model constrains the signal to lie in a *learned k -dimensional subspace*, rather than fixing it to a single vector.

Formal problem statement. Let $x \in \mathbb{C}^d$ be a single observation with noise model $n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$. Given a basis matrix $U \in \mathbb{C}^{d \times k}$ with $k \ll d$, the rank- k constrained ML problem is

$$\min_{U, c} \chi^2(c; U) = (x - Uc)^{\dagger} \Sigma^{-1} (x - Uc), \quad \text{subject to} \quad U^{\dagger} \Sigma^{-1} U = I_k, \quad (42)$$

where $c \in \mathbb{C}^k$ is the coordinate vector of the event in the signal subspace. Over N events with data matrix $X \in \mathbb{C}^{N \times d}$ (rows = events), the joint objective is

$$\min_{U, C} \mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^N (x_i - U c_i)^{\dagger} \Sigma^{-1} (x_i - U c_i), \quad U^{\dagger} \Sigma^{-1} U = I_k, \quad (43)$$

where $C \in \mathbb{C}^{k \times N}$ collects the per-event coefficient vectors.

Structure of the problem. Equation (43) is *biconvex*: for fixed U , the optimal C is obtained by solving N independent linear systems (the E-step); for fixed C , the optimal U is obtained by a weighted regression followed by re-orthonormalization (the M-step). Neither step is a convex problem in the joint variable (U, C) , but the alternating-minimization structure ensures monotone decrease of the objective, as shown in Section 6.

Whitened form. Setting $\tilde{x}_i = \Sigma^{-1/2} x_i$ and $\tilde{U} = \Sigma^{-1/2} U$, the constraint becomes $\tilde{U}^{\dagger} \tilde{U} = I_k$ and the objective (43) reduces to

$$\mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^N \|\tilde{x}_i - \tilde{U} c_i\|_2^2 = \frac{1}{N} \|\tilde{X} - C^{\dagger} \tilde{U}^{\dagger}\|_F^2, \quad (44)$$

a standard low-rank matrix approximation problem in whitened space. Its minimizer is the truncated SVD of $\tilde{X}$, i.e. the top- k right singular vectors of $\tilde{X}$, or equivalently the top- k eigenvectors of the whitened sample covariance $\tilde{X}^{\dagger} \tilde{X} / N = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}$.

Why k matters: signal-family geometry. The optimal rank k^* is the intrinsic dimension of the signal family. When $k < k^*$, the subspace is too small to capture all signal variation; the residual χ^2 contains signal power and the coefficient estimate is biased. When $k > k^*$, extra basis vectors span noise directions; the objective decreases trivially but the additional components carry no signal information. Section 6 discusses criteria for choosing k from data.

4.2 EMPCA algorithm

Following Bailey, let the mean-subtracted dataset be

$$\mathbf{X} = [\mathbf{X}_{:,1}, \dots, \mathbf{X}_{:,M}] \in \mathbb{C}^{N \times M}, \quad \frac{1}{M} \sum_{j=1}^M \mathbf{X}_{:,j} = \mathbf{0}. \quad (45)$$

Classic PCA seeks a rank- K model

$$\mathbf{X} \approx \mathbf{U}\mathbf{C}, \quad \mathbf{U} \in \mathbb{C}^{N \times K}, \quad \mathbf{C} \in \mathbb{C}^{K \times M}, \quad \mathbf{U}^\dagger \mathbf{U} = \mathbf{I}_K, \quad (46)$$

by minimizing the reconstruction error

$$\chi_{\text{PCA}}^2 = \|\mathbf{X} - \mathbf{U}\mathbf{C}\|_F^2 = \sum_{j=1}^M \|\mathbf{X}_{:,j} - \mathbf{U}c_j\|_2^2. \quad (47)$$

For fixed $\mathbf{U}$, the least-squares coefficients are

$$c_j = \mathbf{U}^\dagger \mathbf{X}_{:,j}, \quad \mathbf{C} = \mathbf{U}^\dagger \mathbf{X}. \quad (48)$$

Substituting back gives

$$\chi_{\text{PCA}}^2 = \|\mathbf{X}\|_F^2 - \|\mathbf{U}^\dagger \mathbf{X}\|_F^2, \quad (49)$$

so minimizing χ_{PCA}^2 is equivalent to maximizing captured variance. Therefore $\mathbf{U}$ is given by the top- K eigenvectors of the sample covariance

$$\mathbf{R}_x = \frac{1}{M} \mathbf{X}\mathbf{X}^\dagger. \quad (50)$$

For each event, the reconstruction is $\hat{\mathbf{X}}_{:,j} = \mathbf{U}c_j$; if the global mean was removed, add $\boldsymbol{\mu}$ back after reconstruction.

Statistical optimality and noise assumptions

It is important to emphasize that *standard* PCA admits a maximum-likelihood (ML) interpretation only under a very restrictive noise model. Specifically, consider the generative model

$$\mathbf{x} = \mathbf{s} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma). \quad (51)$$

Standard PCA corresponds to ML estimation of a low-dimensional signal subspace only when the noise covariance is proportional to the identity,

$$\Sigma = \sigma^2 \mathbf{I}. \quad (52)$$

In this case, minimizing the Euclidean reconstruction error is equivalent to minimizing the negative log-likelihood.

Any deviation from this assumption—such as colored noise, frequency-dependent noise power spectral density, or channel-dependent noise variances—breaks the optimality of standard PCA. Under such conditions, Euclidean-distance-based PCA no longer corresponds to a statistically optimal estimator.

Expectation-maximization PCA (EMPCA) restores maximum-likelihood optimality by explicitly incorporating the inverse noise covariance Σ^{-1} into the objective function.

Bailey’s EMPCA formulation solves the same low-rank problem iteratively and avoids explicitly forming the full covariance matrix when N is large. Given current basis $\mathbf{U}^{(r)}$, iteration $r \rightarrow r + 1$ alternates:

E-step (solve coefficients). For each observation $\mathbf{X}_{:,j}$,

$$c_j^{(r+1)} = \arg \min_c \|\mathbf{X}_{:,j} - \mathbf{U}^{(r)} c\|_2^2 = ((\mathbf{U}^{(r)})^\dagger \mathbf{U}^{(r)})^{-1} (\mathbf{U}^{(r)})^\dagger \mathbf{X}_{:,j}. \quad (53)$$

If $\mathbf{U}^{(r)}$ is orthonormal, this reduces to $c_j^{(r+1)} = (\mathbf{U}^{(r)})^\dagger \mathbf{X}_{:,j}$.

M-step (solve basis). With coefficients fixed,

$$\mathbf{U}^{(r+1)} = \arg \min_{\mathbf{U}} \|\mathbf{X} - \mathbf{U} \mathbf{C}^{(r+1)}\|_F^2 = \mathbf{X} (\mathbf{C}^{(r+1)})^\dagger (\mathbf{C}^{(r+1)} (\mathbf{C}^{(r+1)})^\dagger)^{-1}. \quad (54)$$

After each M-step, columns of $\mathbf{U}^{(r+1)}$ are re-orthonormalized (e.g. QR), and iterations continue until $\Delta\chi^2$ or parameter updates are below tolerance.

Equivalent to Bailey’s one-component deflation view, one may solve components sequentially and update

$$\mathbf{X} \leftarrow \mathbf{X} - u_m c_m, \quad (55)$$

before solving the next component, where u_m is the m th column of $\mathbf{U}$ and c_m is the m th row of $\mathbf{C}$.

Yu's microcalorimeter setting motivates a frequency-domain variant that is robust to random time shifts and non-white noise. For event j in Fourier space,

$$X_k^{(j)} = A_j e^{-i\omega_k t_{0,j}} S_k + N_k^{(j)}. \quad (56)$$

Direct PCA on $\{X_k^{(j)}\}$ is problematic because the effective coefficient $A_j e^{-i\omega_k t_{0,j}}$ depends on frequency bin k .

Following Yu, write $X_k^{(j)} = |X_k^{(j)}| e^{i\phi_k^{(j)}}$ and use phase differences

$$\Delta\phi_k^{(j)} = \phi_k^{(j)} - \phi_{k-1}^{(j)}. \quad (57)$$

Under a time shift, $\phi_k^{(j)} \mapsto \phi_k^{(j)} + \omega_k t_{0,j}$, hence

$$\Delta\phi_k^{(j)} \mapsto \Delta\phi_k^{(j)} + (\omega_k - \omega_{k-1}) t_{0,j}. \quad (58)$$

For uniform frequency spacing $\Delta\omega = \omega_k - \omega_{k-1}$, the shift term is independent of k , so the transformed coefficients behave as a single complex factor per event. This restores a linear subspace model in transformed frequency space:

$$\mathbf{X} \approx \mathbf{U}\mathbf{C}. \quad (59)$$

Here $\mathbf{X}$ denotes the transformed (phase-difference) frequency-domain data matrix.

To handle colored/heteroskedastic noise, use Bailey-style weighted EM-PCA with Yu's weighted likelihood:

$$\chi_{\text{EMPCA}}^2 = \sum_{k=1}^N \sum_{j=1}^M (\mathbf{W}_{\text{EMPCA}})_{k,j} |(\mathbf{X})_{k,j} - [\mathbf{U}\mathbf{C}]_{k,j}|^2. \quad (60)$$

Let $\mathbf{W}_j = \text{diag}(w_{1j}, \dots, w_{Nj})$ for column j . The weighted E-step is

$$c_j = \arg \min_c \|\mathbf{W}_j^{1/2} (\mathbf{X}_{:,j} - \mathbf{U}c)\|_2^2, \quad (61)$$

which yields normal equations

$$(\mathbf{U}^\dagger \mathbf{W}_j \mathbf{U}) c_j = \mathbf{U}^\dagger \mathbf{W}_j \mathbf{X}_{:,j}. \quad (62)$$

For the M-step, with $\mathbf{C}$ fixed, each frequency row $\mathbf{X}_{k,:}$ solves

$$u_k^\dagger = \arg \min_{u^\dagger} \|\mathbf{W}_k^{1/2} (\mathbf{X}_{k,:} - u^\dagger \mathbf{C})\|_2^2, \quad (63)$$

with $\mathbf{W}_k = \text{diag}(w_{k1}, \dots, w_{kM})$, yielding

$$(\mathbf{C}\mathbf{W}_k\mathbf{C}^\dagger)u_k = \mathbf{C}\mathbf{W}_k\mathbf{X}_{k,:}^\dagger. \quad (64)$$

In the frequency domain, the EMPCA objective takes the form

$$\chi^2 = \sum_f \left| x(f) - \sum_{m=1}^K c_m u_m(f) \right|^2 \frac{1}{J(f)}, \quad (65)$$

where $J(f)$ is the noise power spectral density.

This expression makes explicit that EMPCA performs subspace estimation in a whitened space. Equivalently, one may view EMPCA as performing PCA after whitening the data by $\Sigma^{-1/2}$. Both interpretations are mathematically identical and correspond to a dual PCA formulation with diagonal noise covariance.

This is the Bailey weighted normal equation in row form and is equivalent to Yu's E/M linear systems in transformed frequency space.

This section matches the common structure in Yu and Bailey: transform to a linear representation, fit a low-rank model, and solve coefficients/basis by alternating weighted least-squares updates.

Dual and frequency-domain interpretation

PCA admits an equivalent *dual* formulation in which the optimization is expressed entirely in terms of inner products. This perspective is particularly natural in the frequency domain.

For stationary noise, the covariance matrix Σ is diagonal in the Fourier basis, with diagonal entries given by the noise power spectral density $J(f)$. In this basis, whitening corresponds to diagonalization of Σ and rescaling by $\Sigma^{-1/2}$.

From this viewpoint, frequency-domain PCA can be interpreted as dual PCA expressed in the Fourier basis. When noise between frequency bins is uncorrelated, the inverse covariance Σ^{-1} acts element-wise in frequency space. As a result, weighted inner products reduce to Hadamard (element-wise) products.

Therefore, the Hadamard-product implementation used in this work is not an approximation or heuristic, but the exact dual-PCA formulation under a diagonal noise covariance.

4.2.1 Gauge freedom and physical amplitude estimation

A key subtlety of EMPCA is that the basis U is defined only up to unitary rotation: if $Q \in \mathbb{C}^{k \times k}$ is unitary, then UQ spans the same subspace as U and achieves the same objective value. Correspondingly, the coefficient vector transforms as $c \rightarrow Q^\dagger c$. The reconstructed signal $\hat{s} = Uc = (UQ)(Q^\dagger c)$ is gauge-invariant, but the individual coefficients $c_1, \dots, c_k$ are not.

For the purposes of energy or amplitude estimation, two gauge-invariant quantities are natural:

1. *Reconstructed signal norm in whitened space.* The squared Mahalanobis norm of the reconstructed signal,

$$\|\hat{s}\|_{\Sigma^{-1}}^2 = \hat{s}^\dagger \Sigma^{-1} \hat{s} = c^\dagger (U^\dagger \Sigma^{-1} U) c = \|c\|_2^2 \quad (\text{under } U^\dagger \Sigma^{-1} U = I_k), \quad (66)$$

is gauge-invariant and measures the total signal power captured by the rank- k model.

2. *Projection onto a calibration direction.* If a fixed reference template s_0 is available, the physical amplitude is estimated as the projection of the reconstructed signal onto s_0 in whitened space:

$$\hat{A} = \frac{s_0^\dagger \Sigma^{-1} \hat{s}}{s_0^\dagger \Sigma^{-1} s_0} = \frac{s_0^\dagger \Sigma^{-1} U c}{s_0^\dagger \Sigma^{-1} s_0}. \quad (67)$$

This reduces to the OF estimator when $k = 1$ and $\text{span}(U) = \text{span}(s_0)$.

4.2.2 Euclidean norm for height estimator

In the frequency-domain setting, the amplitude of the reconstructed signal is often reported as the “height” of the pulse — the peak of the reconstructed time-domain waveform or a scalar proportional to the event energy. When the basis U is orthonormal in whitened space, a simple Euclidean-norm height estimator is available.

Let $\tilde{u}_m = \Sigma^{-1/2} u_m$ denote the m -th whitened basis vector. The EMPCA coefficient vector c satisfies $\hat{c} = \tilde{U}^\dagger \tilde{x}$, where $\tilde{U} = [\tilde{u}_1, \dots, \tilde{u}_k]$. The reconstructed whitened signal is

$$\hat{\tilde{s}} = \tilde{U} \tilde{U}^\dagger \tilde{x}, \quad (68)$$

and its Euclidean norm $\|\hat{s}\|_2 = \|c\|_2$ (since $\tilde{U}^\dagger \tilde{U} = I_k$) is a gauge-invariant measure of captured signal power. To convert to a physical amplitude (e.g. in energy units), one rescales by a calibration factor derived from the expected norm of a reference signal at a known energy, such as a fluorescence X-ray peak.

4.2.3 Ridge regression reconstruction for complex coefficients

In the frequency domain with complex-valued data and a complex basis U , the normal equations for the coefficient vector are

$$(U^\dagger \Sigma^{-1} U) c = U^\dagger \Sigma^{-1} x. \quad (69)$$

When k exceeds the effective rank of $\hat{R}_x$ estimated from the data, or when the basis vectors are nearly collinear in whitened space, the matrix $U^\dagger \Sigma^{-1} U$ becomes ill-conditioned. The standard remedy is Tikhonov (ridge) regularization:

$$\hat{c}_\lambda = (U^\dagger \Sigma^{-1} U + \lambda I_k)^{-1} U^\dagger \Sigma^{-1} x, \quad (70)$$

for a ridge parameter $\lambda > 0$. The resulting estimator is biased toward zero, with bias-variance tradeoff controlled by λ . For real-valued amplitudes with complex frequency-domain representation, one typically takes $\hat{A} = \text{Re}(\hat{c}_1)$ after aligning the first basis vector with the mean pulse direction.

A data-driven choice for λ is the generalized cross-validation (GCV) criterion [16], which minimizes the leave-one-out prediction error. In detector applications where $N \gg k$, the unregularized solve is usually numerically stable after proper re-orthonormalization of U (Section 6.5), and λ can be set to a small numerical floor rather than estimated from data.

4.3 Main Theorem: rank-1 EMPCA equivalent to basic OF

We now formalize the three notions of equivalence used throughout this section.

Estimator equivalence. Two estimators are estimator-equivalent if they produce identical amplitude estimates $\hat{a}_i$ for all events under identical pre-processing, alignment, and noise weighting.

Subspace equivalence. Two methods are subspace-equivalent if they recover the same signal subspace $\mathcal{S} \subset \mathbb{C}^d$, up to trivial gauge freedom. For rank-1 models this means equality up to a nonzero complex scalar; for rank- k models it means equality up to an invertible change of basis within the same subspace.

Amplitude equivalence. Two methods are amplitude-equivalent if, once expressed in the same subspace basis with matched normalization convention, their reconstructed coefficients coincide.

The logical implication relevant here is:

subspace equivalence $\implies$ amplitude equivalence $\implies$ estimator equivalence,

provided the weighting and normalization conventions are matched.

We now state the central equivalence result.

Theorem 4.1 (Rank-1 OF-EMPCA equivalence). *Assume:*

1. *the generative model $x_i = a_i s + n_i$ with fixed template s ;*
2. *Gaussian noise $n_i \sim \mathcal{N}(0, \Sigma)$ with known covariance;*
3. *identical whitening, baseline subtraction, and preprocessing conventions;*
4. *the true signal subspace is one-dimensional.*

Then:

1. *the rank-1 EMPCA subspace equals $\text{span}(\tilde{s})$ in whitened space;*
2. *the resulting EMPCA coefficient estimator equals the generalized least-squares / optimal-filter estimator;*
3. *therefore rank-1 EMPCA and OF are estimator-equivalent under these assumptions.*

Proof. In whitened space,

$$\tilde{x}_i = a_i \tilde{s} + \tilde{n}_i, \quad \tilde{n}_i \sim \mathcal{N}(0, I).$$

The population covariance is

$$\mathbb{E}[\tilde{x}_i \tilde{x}_i^\dagger] = \mathbb{E}[a_i^2] \tilde{s} \tilde{s}^\dagger + I. \tag{71}$$

This is a rank-1 spike perturbing the identity. By standard spiked-covariance theory, the leading eigenvector is proportional to $\tilde{s}$. Hence rank-1 EMPCA recovers

$$u_1 \propto \tilde{s}.$$

Choosing the normalization

$$u_1 = \frac{\tilde{s}}{\|\tilde{s}\|},$$

the EMPCA projection coefficient is

$$\hat{a}_i^{(\text{EMPCA})} = u_1^\dagger \tilde{x}_i = \frac{\tilde{s}^\dagger \tilde{x}_i}{\|\tilde{s}\|}. \quad (72)$$

Restoring the normalization appropriate to the signal model gives

$$\hat{a}_i = \frac{\tilde{s}^\dagger \tilde{x}_i}{\tilde{s}^\dagger \tilde{s}} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}, \quad (73)$$

which is precisely the generalized least-squares / optimal-filter estimator. Therefore subspace equivalence implies amplitude equivalence, which implies estimator equivalence. $\square$

Theorem 4.1 gives the population-level equivalence. The next result states the exact algebraic degeneracy for a fixed basis.

Theorem 4.2 (Exact degeneracy to basic OF). *Assume:*

1. $k = 1$,
2. *the EMPCA basis spans the template direction, i.e. $u = \gamma s_0$ for some $\gamma \neq 0$,*
3. *OF and EMPCA use the same weighting Σ^{-1} ,*
4. *preprocessing and alignment conventions are matched.*

Then rank-1 EMPCA and basic OF yield identical reconstructed signals $\hat{s}$, and therefore identical weighted residual objective values.

Proof. For rank-1 EMPCA,

$$\hat{c} = \frac{u^\dagger \Sigma^{-1} x}{u^\dagger \Sigma^{-1} u}. \quad (74)$$

Substitute $u = \gamma s_0$:

$$\hat{c} = \frac{\gamma^* s_0^\dagger \Sigma^{-1} x}{|\gamma|^2 s_0^\dagger \Sigma^{-1} s_0} = \frac{1}{\gamma} \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0} = \frac{\hat{A}_{\text{OF}}}{\gamma}. \quad (75)$$

Hence

$$\hat{s}_{\text{EMPCA}} = \hat{c} u = \frac{\hat{A}_{\text{OF}}}{\gamma} (\gamma s_0) = \hat{A}_{\text{OF}} s_0 = \hat{s}_{\text{OF}}. \quad (76)$$

Therefore

$$(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$$

is identical for both estimators.

The only remaining freedom is the rank-1 gauge transformation

$$u \mapsto \alpha u, \quad c \mapsto c/\alpha,$$

which can change coefficient convention but not the reconstructed signal. $\square$

4.4 Proposition: χ^2 improvement for $k > 1$

The previous subsections establish exact degeneracy in the rank-1 limit. The main advantage of EMPCA appears when the signal family is not truly one-dimensional. In whitened coordinates, OF is projection onto a 1D subspace $\mathcal{S}_1$, whereas rank- k EMPCA is projection onto a k -dimensional subspace $\mathcal{U}_k$:

$$\hat{\tilde{x}}_{\text{OF}} = P_{\mathcal{S}_1} \tilde{x}, \quad \hat{\tilde{x}}_k = P_{\mathcal{U}_k} \tilde{x}. \quad (77)$$

If $\mathcal{S}_1 \subseteq \mathcal{U}_k$, then by standard projection theory

$$\|\tilde{x} - P_{\mathcal{U}_k} \tilde{x}\|_2^2 \leq \|\tilde{x} - P_{\mathcal{S}_1} \tilde{x}\|_2^2, \quad (78)$$

equivalently

$$\chi_{\text{EMPCA},k}^2 \leq \chi_{\text{OF}}^2. \quad (79)$$

This inequality is purely geometric but has a clear physical interpretation: when the true signal family contains pulse-shape variation, time jitter, channel-coupling modes, or mild nonlinearity, a higher-dimensional subspace can fit the data more accurately than a single fixed template. EMPCA therefore strictly generalizes OF: it reproduces OF exactly in the rank-1 case and improves the fit only when additional physically relevant directions are present.

4.5 Geometric interpretation

The equivalence becomes most transparent after whitening. Using (2), define

$$\tilde{U} = \Sigma^{-1/2}U.$$

Then

$$\chi^2(c) = \|\tilde{x} - \tilde{U}c\|_2^2. \quad (80)$$

Thus both OF and EMPCA are orthogonal projections in whitened coordinates.

For OF, the reconstructed trace is the projection of $\tilde{x}$ onto the one-dimensional subspace $\text{span}(\tilde{s}_0)$:

$$\hat{x}_{\text{OF}} = P_{\tilde{s}_0}\tilde{x}, \quad P_{\tilde{s}_0} = \frac{\tilde{s}_0\tilde{s}_0^\dagger}{\tilde{s}_0^\dagger\tilde{s}_0}. \quad (81)$$

For EMPCA, the reconstructed trace is the projection onto the k -dimensional subspace $\text{span}(\tilde{U})$:

$$\hat{x}_{\text{EMPCA}} = P_{\tilde{U}}\tilde{x}, \quad P_{\tilde{U}} = \tilde{U}(\tilde{U}^\dagger\tilde{U})^{-1}\tilde{U}^\dagger. \quad (82)$$

This immediately explains the hierarchy:

- OF is a projection onto a fixed rank-1 template direction;
- EMPCA is a projection onto a learned rank- k signal subspace;
- when $k = 1$ and the learned subspace equals the template direction, the two estimators coincide exactly.

4.6 Finite-sample considerations and limitation

Finite-sample regime. Theorem 4.1 and the geometric analysis of this section are population-level results: they assume that the sample covariance $\hat{R}_x = N^{-1} \sum_i x_i x_i^\dagger$ equals the population covariance $\mathbb{E}[xx^\dagger] = \mathbb{E}[a^2]ss^\dagger + \Sigma$. In finite samples, $\hat{R}_x$ deviates from its expectation by terms of order $O(d/\sqrt{N})$ in operator norm, which means the estimated EMPCA subspace deviates from the true signal subspace by an angle of order $O(\sqrt{d/N}/\Delta_\lambda)$, where

$$\Delta_\lambda = \lambda_1 - \lambda_2 \quad (83)$$

is the *spectral gap* between the first (signal) and second (noise) eigenvalues of $\Sigma^{-1}\hat{R}_x$. For the rank-1 model, $\Delta_\lambda \approx \mathbb{E}[a^2]\|s\|_{\Sigma^{-1}}^2 = N_\Phi \mathbb{E}[a^2]$, i.e. the spectral gap grows with the signal-to-noise ratio.

Remark 4.1 (Davis–Kahan bound). *By the Davis–Kahan theorem [17], the angle θ between the estimated and true rank-1 EMPCA direction satisfies*

$$\sin \theta \leq \frac{\|\hat{R}_x - R_x\|_{\text{op}}}{\Delta_\lambda}, \quad (84)$$

where $\|\cdot\|_{\text{op}}$ is the operator norm of the estimation error. For Gaussian noise, $\|\hat{R}_x - R_x\|_{\text{op}} = O(\sqrt{d/N})$ with high probability, confirming the scaling above. This bound explains the observed near-unity weighted cosine $\rho_w(U, s) \approx 0.9999999655$ in Table 2: the spectral gap is large (high SNR) and N is large, so the sample subspace is extremely close to the population subspace.

Limitation: algorithmic definition motivates the AE framing. A practical limitation of EMPCA as defined above is that it is an *algorithmic* procedure — it is defined by alternating E- and M-steps — rather than as the minimizer of an explicitly stated objective. This makes it difficult to: (i) state convergence guarantees for the objective value directly, (ii) compare EMPCA with other estimators at the level of the loss function, and (iii) systematically replace the linear encoder with a nonlinear one while preserving the noise-aware structure of the loss.

The noise-aware linear autoencoder formulation of Section 5 resolves all three limitations. By writing the AE objective explicitly as a function of the decoder basis P and showing (via the Bridge Theorem) that its minimizer coincides with the EMPCA subspace, we obtain an objective-first description of the same estimator. This reframing is not merely cosmetic: it is the foundation for replacing the linear decoder with a nonlinear neural network in the companion paper, while retaining the noise-aware Σ^{-1} -weighted reconstruction loss as the training objective.

5 EMPCA as a Noise-Aware Linear Autoencoder

5.1 The encoder-decoder language

5.2 The noise-aware loss

Using the whitening transform of Section 2.3, let $\tilde{X} \in \mathbb{C}^{N \times d}$ denote the whitened data matrix whose rows are $\tilde{x}_i^\dagger = x_i^\dagger \Sigma^{-1/2}$. A tied linear autoen-

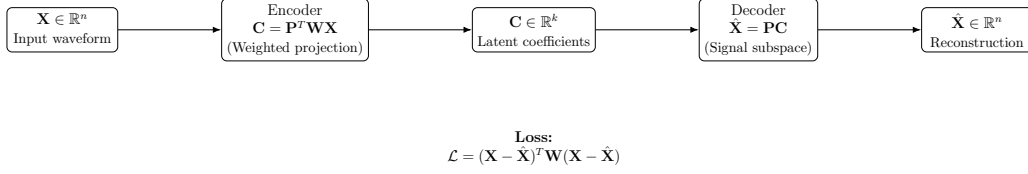


Figure 1: EMPCA as a noise-aware linear autoencoder.

coder with decoder $W \in \mathbb{C}^{d \times k}$, $W^\dagger W = I$, encodes and reconstructs as

$$Z = \tilde{X}W, \quad \hat{\tilde{X}} = ZW^\dagger = \tilde{X}WW^\dagger. \quad (85)$$

The training objective is

$$\min_{W^\dagger W = I} \|\tilde{X} - \tilde{X}WW^\dagger\|_F^2. \quad (86)$$

5.2.1 Native weighted formulation

The whitened formulation above is elegant but obscures the direct connection to the EMPCA objective. We now state the equivalent *native* form in which the noise covariance enters the loss explicitly, without pre-multiplying the data.

Let $P \in \mathbb{C}^{d \times k}$ be a decoder basis with $P^\dagger \Sigma^{-1} P = I_k$ (i.e. orthonormal in the Σ^{-1} -inner product). The encoder is the corresponding Σ^{-1} -weighted projection:

$$z_i = P^\dagger \Sigma^{-1} x_i \in \mathbb{C}^k, \quad (87)$$

and the reconstruction is

$$\hat{x}_i = P z_i = P P^\dagger \Sigma^{-1} x_i. \quad (88)$$

The noise-aware reconstruction loss over N events is

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^\dagger \Sigma^{-1} (x_i - \hat{x}_i). \quad (89)$$

Expanding and using $P^\dagger \Sigma^{-1} P = I_k$:

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^N x_i^\dagger \Sigma^{-1} x_i - \text{Tr} \left(P^\dagger \Sigma^{-1} \left(\frac{1}{N} \sum_i x_i x_i^\dagger \right) \Sigma^{-1} P \right). \quad (90)$$

Since the first term is independent of P , minimizing $\mathcal{L}(P)$ is equivalent to maximizing

$$\text{Tr}(P^\dagger \Sigma^{-1} \hat{R}_x \Sigma^{-1} P), \quad \hat{R}_x = \frac{1}{N} \sum_{i=1}^N x_i x_i^\dagger,$$

subject to $P^\dagger \Sigma^{-1} P = I_k$. By the Rayleigh–Ritz theorem for the generalized eigenproblem

$$\Sigma^{-1} \hat{R}_x \Sigma^{-1} P = \Sigma^{-1} P \Lambda, \quad (91)$$

the optimum P^* spans the top- k eigenvectors of the Σ^{-1} -weighted covariance operator, which is precisely the EMPCA subspace. This confirms that the native-weighted AE objective and the EMPCA objective are the same optimization problem written in different but equivalent coordinates.

5.3 Bridge Theorem: noise-aware linear AE equivalent to EMPCA

Theorem 5.1 (Linear AE in whitened space equals PCA projector). *The minimizer of $\|\tilde{X} - \tilde{X} W W^\dagger\|_F^2$ under $W^\dagger W = I$ is given by the top- k right singular vectors of $\tilde{X}$.*

Proof. Expand:

$$\|\tilde{X} - \tilde{X} W W^\dagger\|_F^2 = \text{Tr}\left((\tilde{X} - \tilde{X} W W^\dagger)^\dagger (\tilde{X} - \tilde{X} W W^\dagger)\right) \quad (92)$$

$$= \|\tilde{X}\|_F^2 - \text{Tr}(W^\dagger \tilde{X}^\dagger \tilde{X} W). \quad (93)$$

So minimizing reconstruction error is equivalent to maximizing $\text{Tr}(W^\dagger \tilde{X}^\dagger \tilde{X} W)$ with orthonormal columns. By Rayleigh–Ritz, optimum W spans the top- k eigenspace of $\tilde{X}^\dagger \tilde{X}$, i.e. top- k right singular vectors of $\tilde{X}$. $\square$

Connection to EMPCA. EMPCA with weight Σ^{-1} minimizes the same Gaussian ML objective as projection in whitened space. Therefore, at convergence, rank- k EMPCA and rank- k complex linear AE should recover the same subspace (up to unitary rotation/gauge).

5.3.1 Bridge theorem (formal statement)

We now state and prove the paper’s second main theorem. It closes the equivalence chain started in Section 2: not only does OF reduce to rank-1

EMPCA (Theorem 4.1), but EMPCA itself is the minimizer of a noise-aware linear autoencoder objective.

Theorem 5.2 (Bridge Theorem: noise-aware linear AE $\equiv$ EMPCA). *Let $x_i = s_i + n_i \in \mathbb{C}^d$ for $i = 1, \dots, N$, where $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$ with $\Sigma \succ 0$ known. Define the noise-aware linear autoencoder loss*

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^N (x_i - PP^\dagger \Sigma^{-1} x_i)^\dagger \Sigma^{-1} (x_i - PP^\dagger \Sigma^{-1} x_i), \quad (94)$$

where $P \in \mathbb{C}^{d \times k}$ satisfies $P^\dagger \Sigma^{-1} P = I_k$. Then:

1. Every minimizer P^* of $\mathcal{L}(P)$ satisfies $\text{range}(P^*) = \text{range}(\hat{U})$, where $\hat{U}$ is the rank- k EMPCA solution with weight Σ^{-1} , up to a unitary rotation within the k -dimensional subspace.
2. Equivalently, P^* spans the top- k eigenvectors of the generalized eigenproblem $\hat{R}_x \Sigma^{-1} v = \lambda v$, where $\hat{R}_x = N^{-1} \sum_i x_i x_i^\dagger$.
3. The minimum value of $\mathcal{L}$ equals $\text{Tr}(\Sigma^{-1} \hat{R}_x) - \sum_{j=1}^k \lambda_j$, where $\lambda_1 \geq \dots \geq \lambda_k$ are the top- k eigenvalues of $\Sigma^{-1} \hat{R}_x$.

Proof. Step 1: reduction to whitened form. Set $\tilde{x}_i = \Sigma^{-1/2} x_i$ and $W = \Sigma^{-1/2} P$. The constraint $P^\dagger \Sigma^{-1} P = I_k$ becomes $W^\dagger W = I_k$. Since

$$PP^\dagger \Sigma^{-1} x_i = \Sigma^{1/2} W W^\dagger \Sigma^{-1/2} x_i = \Sigma^{1/2} W W^\dagger \tilde{x}_i,$$

the loss (94) becomes

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^N (\tilde{x}_i - W W^\dagger \tilde{x}_i)^\dagger (\tilde{x}_i - W W^\dagger \tilde{x}_i) = \frac{1}{N} \|\tilde{X}(I - W W^\dagger)\|_F^2, \quad (95)$$

where $\tilde{X}$ has rows $\tilde{x}_i^\dagger$. This is exactly the tied linear AE loss in whitened coordinates studied in Section 5.3.

Step 2: solution in whitened space. Expanding (95),

$$\mathcal{L}(P) = \|\tilde{X}\|_F^2 - \text{Tr}(W^\dagger \tilde{X}^\dagger \tilde{X} W).$$

Minimizing over W with $W^\dagger W = I_k$ is equivalent, by the Rayleigh–Ritz theorem, to choosing W as the top- k right singular vectors of $\tilde{X}$, i.e. the top- k eigenvectors of $\tilde{X}^\dagger \tilde{X}/N = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}$.

Step 3: translation back to original coordinates. The optimal $P^* = \Sigma^{1/2}W^*$. The subspace $\text{range}(W^*)$ equals the top- k eigenspace of $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, which is unitarily equivalent to the top- k eigenspace of the generalized problem $\hat{R}_x\Sigma^{-1}v = \lambda v$ (left-multiply by $\Sigma^{1/2}$).

Step 4: connection to EMPCA. At convergence, EMPCA with weight Σ^{-1} minimizes the same weighted reconstruction objective (89); its basis spans the same generalized eigenspace. Hence $\text{range}(P^*) = \text{range}(\hat{U})$ up to unitary rotation in the k -dimensional subspace.

Step 5: minimum value. Let $\lambda_1 \geq \dots \geq \lambda_d$ be the eigenvalues of $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2} = \Sigma^{-1}\hat{R}_x$ (same eigenvalues). Then

$$\mathcal{L}^* = \|\tilde{X}\|_F^2 - \sum_{j=1}^k \lambda_j = \text{Tr}(\Sigma^{-1}\hat{R}_x) - \sum_{j=1}^k \lambda_j,$$

which is the unexplained variance in whitened coordinates. $\square$

Remark 5.1. *The Bridge Theorem is a finite-sample statement: it holds for any $N \geq 1$, not just in the population limit. The population version (replacing $\hat{R}_x$ by $\mathbb{E}[xx^\dagger]$) connects to the asymptotic regime studied in Theorem 4.1.*

Remark 5.2 (Gauge freedom). *The minimizer P^* is unique only up to unitary rotation of its columns. Any P^*Q for $Q \in \mathcal{U}(k)$ spans the same subspace and achieves the same loss. Physical amplitudes (e.g. energy estimates) are gauge-invariant because they depend on the reconstructed signal $\hat{x}_i = P^*P^{*\dagger}\Sigma^{-1}x_i$, which is unchanged by Q .*

5.4 EM as coordinate descent on the AE objective

The Bridge Theorem shows that the EMPCA algorithm and the noise-aware linear AE minimize the same objective. We now make the algorithmic connection explicit: the EMPCA E-step and M-step are coordinate-descent steps on the AE loss (94), alternating between optimizing over the latent codes C (encoder output) and the basis P (decoder).

Parameterization. Write the AE loss (89) jointly in (P, C) , where $C = [c_1, \dots, c_N] \in \mathbb{C}^{k \times N}$ collects per-event latent codes:

$$\mathcal{L}(P, C) = \frac{1}{N} \sum_{i=1}^N (x_i - Pc_i)^\dagger \Sigma^{-1} (x_i - Pc_i). \quad (96)$$

The tied encoder constraint (Section 5.2) means that c_i should equal the optimal coefficient for basis P ; it is not a free parameter but rather an implicit function of P . For the coordinate-descent view, however, it is convenient to treat (P, C) as independent variables and impose the encoder constraint at convergence.

E-step: optimize over C with P fixed. For each event i , minimizing (96) over c_i with P fixed gives the weighted normal equations

$$(P^\dagger \Sigma^{-1} P) c_i = P^\dagger \Sigma^{-1} x_i, \quad (97)$$

with solution

$$\hat{c}_i = (P^\dagger \Sigma^{-1} P)^{-1} P^\dagger \Sigma^{-1} x_i. \quad (98)$$

Under the constraint $P^\dagger \Sigma^{-1} P = I_k$, this simplifies to $\hat{c}_i = P^\dagger \Sigma^{-1} x_i$, matching the encoder formula (87). This is precisely the EMPCA E-step.

M-step: optimize over P with C fixed. With C fixed, minimizing (96) over P (subject to $P^\dagger \Sigma^{-1} P = I_k$) is a weighted regression. Define the weighted data aggregates $G = CC^\dagger/N \in \mathbb{C}^{k \times k}$ and $H = CX\Sigma^{-1}/N \in \mathbb{C}^{k \times d}$. The unconstrained solution is

$$P_{\text{unc}}^\dagger = G^{-1} H \iff P_{\text{unc}} = H^\dagger G^{-1}, \quad (99)$$

followed by re-orthonormalization in the Σ^{-1} -inner product (e.g. by QR decomposition of $\Sigma^{-1/2} P_{\text{unc}}$). This matches the EMPCA M-step exactly.

Monotone decrease and convergence. Because each step (98) and (99) is the global minimizer of the objective with respect to its own block of variables,

$$\mathcal{L}(P^{(r+1)}, C^{(r+1)}) \leq \mathcal{L}(P^{(r+1)}, C^{(r)}) \leq \mathcal{L}(P^{(r)}, C^{(r)}). \quad (100)$$

Since $\mathcal{L} \geq 0$, the sequence $\{\mathcal{L}^{(r)}\}$ converges. Formal convergence analysis is given in Section 6.

5.5 Numerical verification: ablation

From the improved equivalence notebook (complex-whitened primary test):

Metric	Value
Principal-angle cosine (EMPCA vs exact complex AE)	0.9999999967
Principal angle (deg)	0.00463536°
Residual KS statistic D	0.00137678
Residual KS p -value	1.0
Relative mean residual difference	8.17299×10^{-7}

Table 3: Primary EMPCA vs exact complex linear AE equivalence metrics (real K-alpha split).

Diagnostic note. If one compares complex EMPCA to a real-feature rank- k model directly, apparent mismatch can increase (e.g. larger KS gap). This is a representation mismatch, not a contradiction of complex-whitened equivalence.

5.6 Connection to Paper 2

The equivalence claim in this chapter is rigorous for:

- linear subspace models,
- Gaussian noise likelihood,
- matched whitening/weighting and preprocessing conventions,
- identical rank k .

It does not automatically extend to nonlinear autoencoders or mismatched objectives.

6 Convergence and Practical Considerations

6.1 Convergence guarantee

The EMPCA algorithm is an instance of coordinate descent on the biconvex objective (96), and its convergence follows from the general theory of block-coordinate descent.

Theorem 6.1 (Monotone convergence of EMPCA). *Let $\mathcal{L}^{(r)}$ denote the value of the noise-aware objective (43) after iteration r of the EMPCA algorithm. Then:*

1. $\mathcal{L}^{(r+1)} \leq \mathcal{L}^{(r)}$ for all $r \geq 0$ (monotone non-increase).
2. The sequence $\{(U^{(r)}, C^{(r)})\}$ has at least one limit point.
3. Every limit point is a fixed point of the alternating-minimization map, i.e. a stationary point of (43).

Proof sketch. Each E-step minimizes $\mathcal{L}$ over C with U fixed; the closed-form solution (98) is the unique global minimizer of a strictly convex quadratic, so $\mathcal{L}(U^{(r)}, C^{(r+1)}) \leq \mathcal{L}(U^{(r)}, C^{(r)})$. Each M-step minimizes $\mathcal{L}$ over U (followed by re-orthonormalization, which cannot increase the objective because the constrained minimizer has weakly smaller objective than the unconstrained one when the orthonormality constraint is active). Thus $\mathcal{L}(U^{(r+1)}, C^{(r+1)}) \leq \mathcal{L}(U^{(r)}, C^{(r+1)}) \leq \mathcal{L}(U^{(r)}, C^{(r)})$. Since $\mathcal{L} \geq 0$, the monotone decreasing sequence converges. Limit-point stationarity follows from continuity of the update map; see Appendix C for full details. $\square$

Remark 6.1 (Global vs local optima). *The objective (43) is non-convex jointly in (U, C) for $k \geq 2$: it has multiple stationary points corresponding to different choices of k eigenvectors from the same eigenspace. However, all stationary points of the rank- k problem correspond to the same k -dimensional eigenspace of $\Sigma^{-1}\hat{R}_x$ (up to degenerate eigenvalues), so the reconstructed signal $\hat{s} = U\hat{c}$ is the same at every stationary point. The global minimum is therefore essentially unique, and in practice EMPCA converges to it from almost any initialization.*

Remark 6.2 (Convergence rate). *Coordinate descent on a biconvex quadratic converges at a linear rate when the signal subspace is well separated from the remaining eigenspace of $\Sigma^{-1}\hat{R}_x$. The asymptotic convergence factor is approximately λ_{k+1}/λ_k , the ratio of the $(k+1)$ -th to the k -th eigenvalue. When this ratio is close to 1 (eigenvalue degeneracy), convergence slows and more iterations are required.*

6.2 Noise modeling assumptions

The framework developed in this paper rests on three assumptions about the noise. We state them precisely and discuss what happens when they are

violated.

Gaussianity. The noise model $n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$ is required for the χ^2 objective to be the negative log-likelihood. Under non-Gaussian noise, minimizing the χ^2 objective still produces a consistent estimator of the signal (by the Gauss–Markov theorem for GLS), but it is no longer the maximum-likelihood estimator and may not be efficient. Heavy-tailed noise (e.g. vibration transients or cosmic-ray glitches) makes the Σ^{-1} -weighted estimator sensitive to outliers. Robust alternatives (e.g. M -estimators with Huber loss) are outside the scope of the present paper.

Stationarity. When noise is wide-sense stationary, Σ is circulant and diagonalizes in the Fourier basis (Section 2.3). Non-stationary noise (e.g. noise that depends on the trigger position, temperature drift, or time-varying electronics) breaks this structure. A practical remedy is to estimate Σ locally in time using a rolling window, or to treat non-stationarity as a modeling error that inflates the effective covariance. Full treatment of non-stationary noise estimation is deferred to the companion paper.

Linearity of the signal model. The rank- k signal model assumes that the true signal s_i lies in a k -dimensional *linear* subspace of $\mathbb{C}^d$. This holds approximately for small variations around a nominal pulse shape, but fails for large timing shifts (where the nonlinear shift manifold is only approximately linear), amplitude saturation (where the detector response is nonlinear), or pulse pile-up. The linear AE bridge (Section 5) makes explicit that relaxing the linear encoder to a nonlinear map — the subject of the companion paper — is the natural next step when these approximations break down.

6.3 Practical initialization

Because the EMPCA objective is biconvex rather than globally convex, different initializations can in principle converge to different stationary points, though in practice the differences are minor (see the remark above on global uniqueness). Three initialization strategies are commonly used.

Random initialization. Draw $U^{(0)}$ as a random orthonormal matrix (e.g. from the Haar measure on the Stiefel manifold) and compute $C^{(0)}$ by the E-

step. This is the simplest approach and works well when $N \gg kd$. Multiple random restarts can be used to confirm convergence to the same subspace.

SVD-based initialization. Compute the top- k right singular vectors of the unweighted data matrix X and use them as $U^{(0)}$. This is equivalent to standard PCA and provides a good warm start when the noise is not too far from isotropic. In whitened coordinates, this corresponds to the top- k eigenvectors of $\hat{R}_x/\sigma^2$ for some scale σ^2 .

Template-seeded initialization. When a fixed template s_0 is available (e.g. from a prior calibration run or a physics-based prediction), set $U^{(0)} = s_0/\|s_0\|_{\Sigma^{-1}}$ as the rank-1 seed and optionally extend to rank k by adding random directions orthogonal to s_0 in whitened space. This initialization is particularly effective for the $k = 1$ case, where it ensures rapid convergence to the OF-equivalent solution, and provides a physically interpretable starting point for $k > 1$.

6.4 Model order selection

The rank k controls the tradeoff between flexibility (ability to capture signal variation) and parsimony (resistance to overfitting noise). Three complementary approaches are standard.

Scree plot and eigenvalue gap. Plot the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of the whitened sample covariance $\Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}$. The signal eigenvalues $\lambda_1, \dots, \lambda_{k^*}$ should be visibly elevated above the noise floor (which, for Gaussian noise with covariance Σ , has expected eigenvalue 1 in whitened coordinates). The “elbow” in the scree plot estimates k^* . For a more formal criterion, look for the largest multiplicative gap: $k^* = \arg \max_j \lambda_j/\lambda_{j+1}$.

Information criteria. For the Gaussian model (42), the AIC and BIC penalize model complexity as

$$\text{AIC}(k) = -2 \log \hat{\mathcal{L}}_k + 2p_k, \quad \text{BIC}(k) = -2 \log \hat{\mathcal{L}}_k + p_k \log N, \quad (101)$$

where $\hat{\mathcal{L}}_k$ is the maximized likelihood under rank- k EMPCA and $p_k = 2kd - k^2$ is the effective number of free real parameters in the orthonormal basis U (accounting for the k^2 parameters absorbed by the unitary gauge freedom).

The BIC tends to select smaller k than AIC, which is preferable when N is moderate.

Cross-validated chi-squared. Hold out a fraction of events (or a fraction of frequency bins) and evaluate the out-of-sample χ^2 as a function of k . The optimal k is where the out-of-sample χ^2 stops decreasing. This is the most reliable criterion but requires splitting the dataset and is more expensive to compute. For detector calibration studies where N is large, cross-validation is recommended.

Remark 6.3. *In the cryogenic detector setting, a natural physical upper bound on k is the number of physically distinct pulse-shape variation modes (e.g. number of independent timing or geometry parameters). Incorporating this prior information can dramatically narrow the search range for k .*

6.5 Conditioning and regularization

Two numerical issues arise in practice: ill-conditioned noise PSDs and ill-conditioned basis matrices.

Ill-conditioned noise covariance. The whitening step requires computing $\Sigma^{-1/2}$ or dividing by $\sqrt{J_k}$ at each frequency. If $J_k \approx 0$ for some bins (e.g. above the detector bandwidth or at DC), the corresponding whitened amplitudes are very large and amplify numerical errors. Three regularization strategies:

1. *Additive floor.* Replace $J_k \rightarrow \max(J_k, \varepsilon)$ for a floor $\varepsilon = \delta \cdot \max_k J_k$ with $\delta \in [10^{-4}, 10^{-2}]$ chosen by inspection of the PSD.
2. *Frequency masking.* Zero out bins outside a pre-defined analysis band $[f_{\min}, f_{\max}]$ before whitening. This is equivalent to projecting onto the informative subspace and is lossless when the signal has negligible power outside the band.
3. *Tikhonov regularization of Σ^{-1} .* Replace $\Sigma^{-1} \rightarrow (\Sigma + \varepsilon I)^{-1}$ for a ridge parameter $\varepsilon > 0$. This shrinks the whitened inner product at all frequencies uniformly and is appropriate when the noise floor is uncertain.

Ill-conditioned basis matrix. When the EMPCA basis vectors $u_1, \dots, u_k$ are nearly collinear in whitened space (e.g. because k exceeds the effective dimension of the signal family), the matrix $U^\dagger \Sigma^{-1} U$ becomes ill-conditioned and the coefficient solve (98) is numerically unstable. Remedies:

1. *Re-orthonormalization.* Re-orthonormalize U at every M-step (QR decomposition of $\Sigma^{-1/2} U$) to maintain $U^\dagger \Sigma^{-1} U = I_k$. This is standard in all numerical EMPCA implementations.
2. *Ridge on the coefficient solve.* Replace $(U^\dagger \Sigma^{-1} U)^{-1}$ by $(U^\dagger \Sigma^{-1} U + \lambda I_k)^{-1}$ for a small $\lambda > 0$. This biases the coefficients toward zero but prevents explosion of $\hat{c}_i$ when the basis is nearly degenerate. The optimal λ can be chosen by cross-validation or by monitoring the condition number of $U^\dagger \Sigma^{-1} U$.
3. *Reducing rank.* If the condition number of $U^\dagger \Sigma^{-1} U$ is persistently large despite re-orthonormalization, this is a sign that k exceeds the effective signal dimension. Reduce k by one and re-run.

7 Experimental Validation

7.1 Benchmark domain

The purpose of Study A is to determine which noise source dominates in each frequency band, where crossover frequencies occur, and whether the detector is readout-limited or device-limited in the relevant regimes.

7.2 Evaluation metrics

7.3 Verification of equivalence theorems

7.4 Reconstruction quality versus rank k

Study B addresses a different question: which noise source most strongly limits baseline energy resolution. This is not necessarily the visually largest source in the ASD plot, because optimal filtering weights the signal and noise spectrally.

Table 4: Predicted relative sensitivity for leave-one-out and scale-down scenarios.

Scenario	Relative sensitivity	Improvement vs baseline
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7.4.1 Signal definition and OF weighting

Let $S(f)$ denote the frequency-domain signal template. The inverse-variance weighting relevant for optimal filtering is governed by

$$W(f) = \frac{|S(f)|^2}{P_{\text{tot}}(f)}. \quad (102)$$

The cumulative information content and the effective signal band are then derived from $W(f)$.

7.4.2 Baseline sensitivity and leave-one-out study

Using the total PSD, we compute a baseline sensitivity proxy

$$I \propto \int df \frac{|S(f)|^2}{P_{\text{tot}}(f)}, \quad \sigma_E \propto I^{-1/2}, \quad (103)$$

up to the calibration convention of $S(f)$. We then recompute the sensitivity after removing one component at a time.

7.4.3 Component scale-down study

To rank realistic engineering priorities, each component is additionally scaled by factors such as 0.8, 0.5, and 0.2, and the resulting improvement in σ_E is evaluated.

7.4.4 Study B summary

Study B translates the qualitative Study A noise-budget picture into a quantitative ranking of which hardware or readout improvement is expected to yield the largest reconstruction benefit.

[Figure placeholder: studyB_weighting_density]

Figure 2: Optimal-filter weighting density used in Study B.

7.5 Noise-aware loss versus isotropic MSE

7.6 Convergence behavior

whitenoise, MMC noise and pink noise

7.6.1 Computation Enhancement of PCA

1. Sliding-window inference (for the next paper): fixed eigenvectors with fast coefficient solves; output amplitude, χ^2 , and likelihood traces.
2. Optimizing EMPCA with cuBLAS / batched GEMM: higher ROI for acceleration. Example targets include batched linear solves ($Ac = b$), fused weighting/projection/reduction kernels, and improved data-flow/latency management.

8 Discussion and Conclusion

8.1 The noise-aware loss as a transferable principle

The estimator hierarchy of Table 1 is built on a single organizing principle: a reconstruction method is statistically efficient when and only when its training objective respects the geometry imposed by the noise covariance. Changing the architecture while keeping an isotropic loss produces a more expressive model that is nonetheless suboptimal; changing the loss to encode Σ^{-1} while keeping a simple architecture recovers optimality. This asymmetry — that the loss is the load-bearing component, not the architecture — is the transferable lesson of the linear theory.

This principle manifests at every level of the estimator hierarchy. Optimal filtering is statistically efficient for a fixed template precisely because it weights the data by Σ^{-1} : it down-weights frequency bins with large noise PSD and up-weights bins where the signal is informative. Standard PCA achieves the same efficiency only under isotropic noise ($\Sigma \propto I$); when noise is colored or heteroskedastic, PCA uses the wrong geometry and is no longer the ML estimator. EMPCA corrects this by incorporating Σ^{-1} into the E- and M-step updates. The noise-aware linear AE makes the same correction at the level of the training loss: replacing Euclidean MSE by the Σ^{-1} -weighted reconstruction error is sufficient to recover the EMPCA solution without any change to the linear encoder–decoder architecture.

This observation has practical implications for the broader ML community. In scientific instrument settings — calorimeters, gravitational-wave detectors, radio telescopes, medical imaging arrays — the noise model is often well characterized from dedicated calibration runs, yet reconstruction networks are routinely trained with isotropic MSE or binary cross-entropy losses that ignore this information. The framework presented here makes the cost explicit: using isotropic MSE in the presence of structured noise is equivalent to whitening the residuals incorrectly, which can substantially degrade reconstruction quality (see Section 7.4 and the noise-aware vs MSE ablation in Section 7). Replacing MSE by the noise-aware weighted loss requires no architectural change; it is a targeted, principled upgrade that should be the default starting point for reconstruction tasks with known noise statistics.

8.2 Relation to PPCA, factor analysis, and VAE

The framework developed in this paper sits at the intersection of several well-established model families. We discuss the precise relationships.

Probabilistic PCA (PPCA). (author?) [7] derived PCA as the maximum-likelihood estimator for the latent variable model $x = Wz + \mu + \varepsilon$, with $z \sim \mathcal{N}(0, I)$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, and $W \in \mathbb{R}^{d \times k}$. The key constraint is *isotropic* noise: $\Sigma = \sigma^2 I$. Under this assumption, PPCA recovers the same subspace as standard PCA (scaled by σ), and the EM algorithm of PPCA is algebraically identical to the EMPCA E- and M-steps in whitened coordinates with $\Sigma^{-1} = (1/\sigma^2)I$.

The present paper is the *heteroskedastic generalization* of PPCA: we replace $\sigma^2 I$ by a general $\Sigma \succ 0$ and derive the corresponding ML estimator and EM algorithm. The resulting EMPCA algorithm, the equivalence to OF at rank 1, and the AE bridge theorem are all new relative to the PPCA literature.

Factor analysis. Factor analysis [8] uses the model $x = \Lambda f + \varepsilon$ with $f \sim \mathcal{N}(0, I)$ and $\varepsilon \sim \mathcal{N}(0, \Psi)$, where $\Psi = \text{diag}(\psi_1, \dots, \psi_d)$ is a diagonal but not isotropic noise covariance. Factor analysis is thus a partial heteroskedastic generalization of PPCA: it allows different noise variances per feature dimension but does not model inter-feature noise correlations. The present framework goes further by allowing a fully general (non-diagonal) Σ , which is essential for applications with correlated noise across time bins or across channels. Factor analysis and the framework here coincide when Σ happens to be diagonal, which in the frequency domain corresponds to the stationary noise case.

Variational autoencoders (VAE). The VAE [18] uses a *nonlinear* probabilistic encoder $q_\phi(z|x)$ and decoder $p_\theta(x|z)$ trained to maximize the evidence lower bound (ELBO). In the special case of a linear encoder, a Gaussian posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2 I)$, and a Gaussian likelihood $p_\theta(x|z) = \mathcal{N}(Wz, \Sigma_{\text{dec}})$, the VAE ELBO reduces (in the limit $\sigma \rightarrow 0$) to the noise-aware reconstruction loss studied here. The KL regularization term in the ELBO (the prior on z) is absent in our framework because we treat z as a deterministic latent code rather than a random variable. Adding KL regularization to the noise-aware linear AE would recover a linear variant of the

VAE with heteroskedastic observation noise — a connection that may be useful for uncertainty quantification in detector reconstruction.

Denoising autoencoders. (author?) [14] proposed denoising AEs (DAEs), which are trained to reconstruct the clean input from a corrupted version. Under Gaussian corruption with covariance Σ_{corr} , the optimal linear DAE recovers the same subspace as the noise-aware AE with $\Sigma = \Sigma_{\text{corr}}$ [19]. The present framework can therefore be interpreted as a denoising AE where the corruption distribution is the *measurement noise distribution* rather than an artificial training corruption.

8.3 Limitations

We are explicit about the boundaries of the present framework.

Requires known or estimated noise covariance. All estimators in the hierarchy require Σ (or equivalently, the noise PSD J_k) to be known or estimated from dedicated noise data. In most detector applications this is standard practice: noise is estimated from triggered baselines or from periods when no signal is present. However, if the noise is non-stationary (drifting baselines, microphonic bursts, glitch artifacts), the noise estimate may be stale or biased, which degrades the whitening and breaks the optimality guarantees. Robust noise estimation is an active research area and outside the scope of this paper.

Gaussian noise assumption. The χ^2 objective is the correct negative log-likelihood only for Gaussian noise. For distributions with heavier tails (e.g. Poisson photon counting, $1/f$ noise with non-Gaussian fluctuations, or electronic transients), the Gaussian MLE is consistent but not efficient. Robust alternatives require replacing the quadratic loss by a robust M -estimator objective, which breaks the closed-form EM structure and requires iteratively reweighted least squares or gradient methods.

Linear signal model. The rank- k subspace model assumes that the signal family lies in a k -dimensional *linear* subspace. This is an approximation: the true manifold of shifted pulses, position-dependent shapes, and multi-mode excitations is generally nonlinear. For small variations around a nominal

template, the linear approximation is excellent (as demonstrated by the near-unity cosine in Table 2). For large timing shifts or strongly nonlinear detector response, a nonlinear model is required.

Single-pass, offline learning. The present implementation of EMPCA requires all N events to be available simultaneously for the subspace update. Online or streaming variants — useful for real-time triggering and adaptive noise tracking — require extensions that maintain a rank- k subspace estimate as new events arrive. Stochastic gradient methods for PCA [20] and online EM algorithms [21] provide principled starting points, but their adaptation to the noise-aware objective is not developed here.

8.4 Companion paper preview

The present paper establishes that optimal filtering, EMPCA, and a noise-aware tied linear autoencoder are three views of the same linear-subspace constrained Gaussian ML estimator. The linear theory provides a principled baseline: it gives the best possible reconstruction within the class of linear models, and the noise-aware objective is the right starting point for any generalization.

The companion paper (Paper 2) relaxes the linear encoder to a deep neural network while retaining the noise-aware Σ^{-1} -weighted reconstruction loss as the training objective. The key architectural change is replacing $z = P^\dagger \Sigma^{-1} x$ (the linear encoder) by $z = f_\theta(x)$ (a learned nonlinear encoder such as a ResNet or convolutional network), while keeping the decoder linear (or lightly nonlinear) and the loss in the noise-aware form. This design choice — nonlinear encoding with noise-aware loss — is motivated directly by the linear theory: the encoder is relaxed because it is where the linear approximation to the signal manifold breaks down, while the loss is kept noise-aware because that is the component that ensures statistical efficiency.

The companion paper additionally develops sliding-window inference (fixed eigenvectors with fast coefficient updates), GPU acceleration via batched GEMM, and a real-time triggering pipeline that uses the noise-aware χ^2 as a detection statistic.

8.5 Conclusion

We have shown that optimal filtering, expectation–maximization PCA, and a noise-aware tied linear autoencoder are three constrained solutions of a single Gaussian maximum-likelihood objective with explicit covariance weighting. The unifying structure is the whitening transform: after mapping to whitened coordinates, all methods become orthogonal projections that differ only by the constraint imposed on the admissible subspace — rank-1 fixed template (OF), rank-1 shift manifold (time-shift OF), rank-1 joint stacked vector (joint-channel OF), or rank- k learned subspace (EMPCA and noise-aware linear AE).

The main theoretical contributions are two equivalence theorems. Theorem 4.1 proves that, under matched whitening, rank-1 EMPCA recovers the optimal-filter amplitude estimator exactly. Theorem 5.2 (the Bridge Theorem) proves that the minimizer of the noise-aware linear AE loss spans the same subspace as EMPCA at convergence, closing the loop from the signal-processing, statistical, and machine-learning perspectives into a single coherent framework.

The practical message is equally direct: when noise statistics are known or measurable, they should be incorporated into the reconstruction objective rather than absorbed into the model architecture. This principle is architecture-independent: it holds for the classical matched filter, for the statistical EM algorithm, and for the neural network encoder–decoder. The noise-aware loss is the transferable element, and the linear theory developed here defines the baseline against which any nonlinear generalization should be evaluated.

A Proof of Main Theorem

B Proof of Bridge Theorem

C Convergence proof details

D Simulation and dataset details

D.1 LAMCAL and cryogenic detector response

D.2 Quasi-particle signal formation

D.3 Simulation implementation

Summary of Section 1.3: Simulation Implementation

1.3.1 Noise Simulation Implementation. The noise simulation is implemented through a dedicated `NoiseGenerator` class, designed to provide frequency-domain colored noise in a compact and modular way. The workflow separates configuration from generation: spectral definitions and normalization are handled during setup, while each call to the generator produces a time-domain trace consistent with the selected power spectral density (PSD). Built-in models include standard noise colors (white, blue, violet, pink, brownian), each corresponding to a simple analytic frequency dependence. The implementation first normalizes the spectral shape so that it controls only how power is distributed across frequency, and then applies discrete FFT scaling to ensure the total noise power matches the configured level. Noise traces are synthesized in frequency space by assigning magnitudes from the PSD and random phases, followed by an inverse real FFT to obtain the time-domain waveform. A custom PSD mode allows user-supplied spectra to be interpolated and used directly. Overall, the method ensures physically consistent mapping between continuous PSD definitions and discrete sampled waveforms.

1.3.2 TraceSimulator Architecture and Signal-Formation Workflow.

The `TraceSimulator` package follows a modular architecture in which detector response maps, energy partitioning, geometry handling, and noise generation are clearly separated. For each simulated event, deposited recoil energy

is first partitioned into phonon, triplet, UV, and IR components using the `DELIGHTSignalFormation` model. These energy components are converted into discrete quanta and distributed across detector channels according to position-dependent efficiencies derived from response maps. Arrival times are sampled from precomputed transport templates, histogrammed onto digitizer bins, and converted into ADC-equivalent amplitudes.

The quasi-particle (QP) signal is formed primarily through the phonon branch in vacuum channels. The number of phonon quanta is computed from the phonon energy component, scaled by the local phonon collection efficiency (PCE), and fluctuated with Poisson statistics. Arrival times are sampled from the phonon arrival-time distribution, histogrammed in time, scaled by detector gain and energy-to-ADC conversion factors, and finally shaped with the detector pulse template (shifted-sum convolution). Conceptually, the chain is:

Energy $\rightarrow$ Quanta $\rightarrow$ Channel Acceptance (PCE) $\rightarrow$ Arrival-Time Sampling $\rightarrow$ Histogramming $\rightarrow$

After signal construction, colored noise from the `NoiseGenerator` is added, and optional digitization effects are applied. A standard simulator produces baseline waveforms, while an extended long-trace version additionally provides clean/noisy pairs and signal masks for analysis studies. The overall design emphasizes clear separation between physics modeling, geometry handling, and stochastic waveform generation.

Energy-to-Quanta Conversion. After energy partitioning by `DELIGHTSignalFormation`, the deposited recoil energy E is decomposed into four components

$$E = E_{\text{ph}} + E_{\text{tr}} + E_{\text{uv}} + E_{\text{ir}},$$

corresponding to phonon, triplet, UV, and IR channels.

Each component is converted into discrete quanta using the corresponding mean excitation energy scale. In the implementation this is approximated as

$$N_{\text{uv}} = \left\lfloor \frac{E_{\text{uv}}}{E_{\text{UV}}} \right\rfloor, \quad N_{\text{tr}} = \left\lfloor \frac{E_{\text{tr}}}{E_{\text{triplet}}} \right\rfloor,$$

$$N_{\text{ir}} = \left\lfloor \frac{E_{\text{ir}}}{E_{\text{IR,avg}}} \right\rfloor, \quad N_{\text{ph}} \approx \frac{E_{\text{ph}}}{\epsilon_{\text{ph}}},$$

where in the current code convention $\epsilon_{\text{ph}} \simeq 1$ meV per phonon quantum.

Thus, the continuous deposited energy is mapped onto integer excitation numbers, which are the fundamental stochastic objects propagated through the detector model.

Channel Distribution and Detection Statistics. For optical (UV/IR) and triplet channels, quanta are distributed across LAMCAL channels according to position-dependent light-collection efficiencies (LCE) or triplet-collection efficiencies (TCE). This is implemented as a multinomial draw conditioned on the local efficiency vector $\{p_i(x, y, z)\}$ with

$$\sum_i p_i(x, y, z) \leq 1.$$

For phonons (relevant for QP formation in vacuum channels), the expected number of detected phonons in channel i is

$$\lambda_i(x, y, z) = \text{PCE}_i(x, y, z) N_{\text{ph}},$$

where PCE_i is the phonon collection efficiency from the response map. The observed phonon count is then drawn as

$$N_{\text{ph},i}^{\text{obs}} \sim \text{Poisson}(\lambda_i),$$

explicitly modeling counting fluctuations at the quasi-particle level.

Arrival-Time Sampling and Histogram Formation. For each detected quantum, an arrival time is sampled from the precomputed transport distribution

$$t_{\text{ph},i}^{(k)} \sim P_{\text{arr},i}(t \mid x, y, z),$$

obtained from the phonon arrival-time maps. The resulting discrete arrival times are histogrammed into digitizer bins of width $\Delta t = 1/f_s$:

$$H_i[n] = \sum_k \mathbf{1}\left(t_{\text{ph},i}^{(k)} \in [n\Delta t, (n+1)\Delta t)\right).$$

This produces a discrete time-series representation of deposited phonon quanta per channel.

Template Shaping and QP Pulse Formation. The histogrammed counts are converted into ADC-equivalent amplitudes via gain and calibration constants G_i :

$$A_i[n] = G_i H_i[n].$$

Detector response shaping is then applied through convolution with the channel-specific pulse template $T_i[n]$:

$$s_i[n] = (A_i * T_i)[n] = \sum_m A_i[m] T_i[n - m].$$

Conceptually, the QP signal formation chain can therefore be written as

$$E_{\text{ph}} \rightarrow N_{\text{ph}} \rightarrow \lambda_i \rightarrow N_{\text{ph},i}^{\text{obs}} \rightarrow H_i[n] \rightarrow A_i[n] \rightarrow s_i[n].$$

Finally, colored noise $\eta_i[n] \sim \mathcal{N}(0, \Sigma)$ generated by `NoiseGenerator` is added:

$$x_i[n] = s_i[n] + \eta_i[n],$$

optionally followed by digitization to integer ADC counts.

This formalizes the simulation as a sequence of stochastic mappings from continuous recoil energy to discrete quasi-particle statistics, then to deterministic template-shaped waveforms, and finally to noisy measured traces.

D.4 Input spectra and working conventions

We use the uploaded component spectra `total_noise.dat`, `SQUID_noise.dat`, `Johnson_noise.dat`, `TD_noise.dat`, `Er_noise.dat`, and the auxiliary MMC reference spectrum. For the AL spectra, the source figure labels the vertical axis as $\sqrt{S_\phi}$ in units of $\mu\Phi_0/\sqrt{\text{Hz}}$, so these curves are interpreted as amplitude spectral densities (ASD), not PSD. Following the internal working convention $1 \text{ V}/\Phi_0$, we use $\mu\Phi_0 \leftrightarrow \mu\text{V}$ for relative comparison. Thus, AL PSDs are constructed as

$$P_i(f) = N_i^2(f), \tag{104}$$

where $N_i(f)$ denotes the ASD of component i .

Before physics interpretation, we verify:

- common frequency-grid consistency,
- monotonicity and finiteness of the spectra,
- quadrature consistency of the total AL noise.

D.5 Study A: Noise budget decomposition

The purpose of Study A is to determine which noise source dominates in each frequency band, where crossover frequencies occur, and whether the detector is readout-limited or device-limited in the relevant regimes.

D.5.1 Quadrature consistency of the total spectrum

We test whether the total ASD is consistent with quadrature summation of the identified components:

$$N_{\text{model}}^2(f) = N_{\text{Er}}^2(f) + N_{\text{Johnson}}^2(f) + N_{\text{SQUID}}^2(f) + N_{\text{TD}}^2(f). \quad (105)$$

The reconstructed ASD is then

$$N_{\text{model}}(f) = \sqrt{N_{\text{model}}^2(f)}. \quad (106)$$

We compare $N_{\text{model}}(f)$ to the provided total ASD and evaluate their ratio across the native AL frequency grid.

D.5.2 Overview of the component budget

Figure 3 shows the total ASD together with the individual component ASDs. This figure provides the visual overview of the noise budget and reveals the broad dominance regimes.

D.5.3 Fractional variance contributions

To quantify the relative importance of each source, we compute the PSD-based fractional contribution

$$R_i(f) = \frac{P_i(f)}{P_{\text{tot}}(f)} = \frac{N_i^2(f)}{N_{\text{tot}}^2(f)}. \quad (107)$$

A value $R_i(f) \approx 1$ indicates dominance of component i in that frequency band, while comparable values among multiple components indicate a mixed regime.

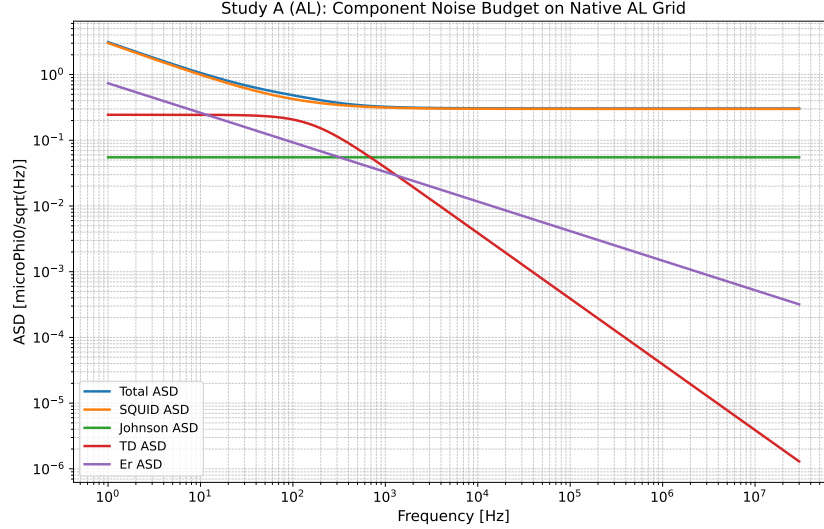


Figure 3: Study A overview of the AL noise budget on the native AL frequency grid.

Table 5: Dominant noise source by frequency band for Study A.

Band	Dominant	Secondary	Dominant fraction	Interpretation
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D.5.4 Dominant source by band and crossover frequencies

We divide the analyzed frequency range into broad logarithmic bands and compute the median fractional contribution of each component in each band. This yields a compact dominant-source summary. We also determine approximate crossover frequencies by solving

$$P_i(f) = P_j(f) \quad (108)$$

for selected component pairs.

D.5.5 Study A summary

Study A provides the validated AL-only noise-budget interpretation. The provisional MMC comparison is kept separate because the physical normalization of the raw MMC spectrum has not yet been independently verified.

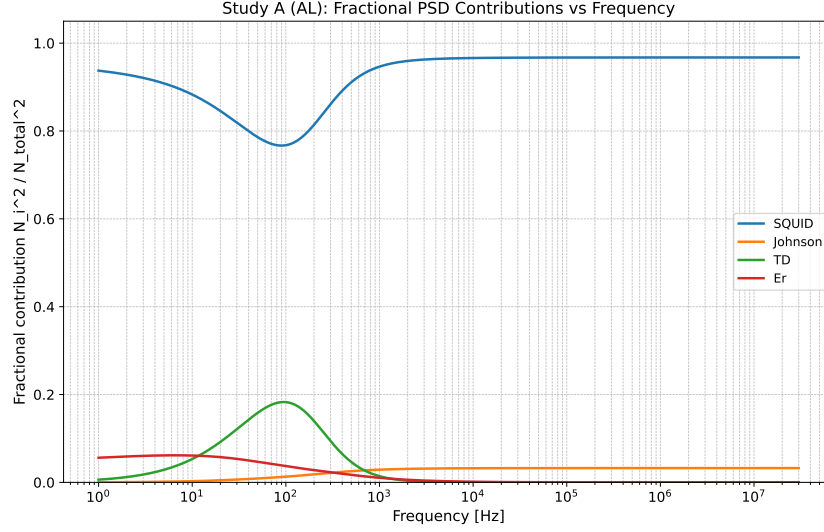


Figure 4: Fractional PSD contributions of the AL noise components versus frequency.

Table 6: Crossover frequencies between selected noise components.

Component pair	Crossover frequency [Hz]	Comment
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D.6 Implementation for the package NoiseGenerator

E List of Symbols

This section summarizes the notation used throughout this work, unifying Optimal Filtering (OF), Principal Component Analysis (PCA), and Expectation–Maximization PCA (EMPCA) in both time and frequency domains.

Category	Symbol	Meaning
Indices	i	Discrete time-sample index within a trace, $i = 0, \dots, N - 1$.
	k	Frequency-bin index of the discrete Fourier transform.
	j	Trace (event) index in a dataset of size M .

Category	Symbol	Meaning
	m	Principal component index, $m = 1, \dots, K$.
Time / frequency	t, ω	Continuous time and angular frequency, $\omega = 2\pi f$.
	$\Delta\omega$	Frequency-bin spacing, $\Delta\omega = \omega_k - \omega_{k-1}$.
Measured data	$x(t), x_i$	Measured detector trace in the time domain.
	$X(f), X_k$	Fourier transform of the measured trace.
	$\mathbf{x}$	Vectorized single-event trace (time or frequency representation, depending on context).
	$\bar{X}_k$	Mean spectrum over events, $\bar{X}_k = \frac{1}{M} \sum_{j=1}^M X_k^{(j)}$.
	$\phi_k, \Delta\phi_k$	Phase of X_k and phase difference $\Delta\phi_k = \phi_k - \phi_{k-1}$ used in shift-invariant frequency transforms.
Signal model	$s(t), s_i$	Signal template (pulse shape) in time domain.
	$S(f), S_k$	Fourier transform of the signal template.
	$\mathbf{s}, \hat{\mathbf{s}}$	Vectorized signal template and reconstructed signal for a single event.
	S_k^{POF}	PCA-assisted OF template built from the aligned first principal component.
	A	Signal amplitude (pulse height).
	t_0	Signal arrival time (time offset).
	α^*	Global phase-alignment parameter used to fix the phase ambiguity of the first principal component.
Noise	$n(t), N(f)$	Additive detector noise in time and frequency domains.
	$\mathbf{n}$	Vectorized additive noise for a single event.

Category	Symbol	Meaning
	Σ	Noise covariance matrix; for stationary noise, Σ is diagonal in frequency space and related to J_k .
	$J(f), J_k$	Noise power spectral density (PSD).
	$W(f)$	Inverse-noise weight, $W(f) = 1/J(f)$.
Optimal Filtering	$\chi^2(A, t_0)$	Weighted least-squares objective defining the OF maximum-likelihood estimator.
	$\Phi_k, \Phi_k^{(0)}$	Unnormalized and normalized optimal-filter kernels.
	$N_\Phi, B(t_0)$	Filter normalization and delayed matched-filter response used in profiled $\chi^2(t_0)$.
	β_0, β_1	Affine calibration coefficients mapping raw estimator output to calibrated amplitude.
Dataset (PCA)	$\mathbf{X} \in \mathbb{C}^{N \times M}$	Dataset matrix whose columns are individual traces.
	$\boldsymbol{\mu}$	Mean trace of the dataset.
	$\tilde{\boldsymbol{\mu}}, \mathbf{R}_{\tilde{s}}$	Whitened mean signal and corresponding covariance matrix used in the PCA/OF equivalence discussion.
PCA / EMPCA	$\mathbf{U} \in \mathbb{C}^{N \times K}$	Matrix of principal components (basis vectors).
	$\mathbf{C} \in \mathbb{C}^{K \times M}$	Coefficient matrix for PCA/EMPCA reconstruction.
	$\hat{\mathbf{X}}$	Low-rank reconstruction, $\hat{\mathbf{X}} = \boldsymbol{\mu} \mathbf{1}^\top + \mathbf{U} \mathbf{C}$.
	$\mathbf{W}_{\text{EMPCA}}$	Element-wise weight matrix used in weighted PCA / EMPCA (distinct from the spectral scalar weight $W(f) = 1/J(f)$).
Multi-channel	$x_c(t), X_{c,k}$	Time / frequency-domain signal for channel c .

Category	Symbol	Meaning
	$J_{n,c}(f)$	Channel-dependent noise PSD.

F Implementation of Optimal Filter

G Implementation of EMPCA

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